ProtMind

Software Requirements Specification

Unified Intelligent Medical Bioinformatics Platform

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Protein Research & AI Bioinformatics

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Chapter 1: Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional and non-functional requirements for ProtMind — a unified, intelligent medical bioinformatics platform designed to consolidate protein research infrastructure into a single coherent system.

The purpose of this document is to provide a comprehensive, unambiguous, and verifiable specification of ProtMind's intended behavior, system architecture, data management strategy, and external interface requirements. It serves as the authoritative reference for all development, testing, validation, and stakeholder communication activities throughout the project lifecycle.

This document covers all seven development phases: data infrastructure, visualization suite construction, MVP (Model 1 — Protein Function Prediction), explainability and trust layer, Model 2 (Protein-Protein Interaction Prediction), Model 3 (Cross-Species Essentiality and Mutation Analysis), and the final CCA clinical validation and demonstration.

1.2 Document Conventions

Convention	Meaning
Bold text	Key terms, system component names, or field names introduced for the first time
Italics	External tool names, database names, or technical terminology
FR-XXX	Functional Requirement identifier (e.g., FR-001 = first functional requirement)
NFR-XXX	Non-Functional Requirement identifier
UC-XXX	Use Case identifier
[PHASE N]	Requirement or feature scoped to a specific development phase
TBD	To Be Determined — requires further research or stakeholder input
API	Application Programming Interface
PDB	Protein Data Bank — repository of experimentally-determined protein structures
GO	Gene Ontology — standardized vocabulary for protein function annotation
CCA	Cholangiocarcinoma — bile duct cancer, used as the anchor validation disease

1.3 Intended Audience and Reading Suggestions

This document is written for multiple audiences with different reading priorities:

Audience	Recommended Sections
Development Team (Backend/ML)	Chapters 3 (FR), 4.1 (Architecture), 4.2 (Stack), 4.5 (ERD), 4.6 (Activity Diagrams)
Frontend / Visualization Engineers	3.1 (UI), 4.3 (Use Cases), 4.8 (Prototype), 4.9 (Tooling)
Data Scientists / ML Engineers	1.4–1.6, 2.2, 3.7, 4.1, 4.2
Research Collaborators & Advisors	Chapters 1 and 2 — overview and scope sections
Clinical / Medical Stakeholders	1.4, 1.5, 2.3, 2.6, Chapter 3 stakeholder analysis
QA / Testing Engineers	Chapter 3 FR/NFR, 4.3 Use Case Specs, 4.6 Activity Diagrams
System Administrators	3.2 (Hardware), 3.3 (Software), 4.1 (Architecture), 4.9 (Tooling)

1.4 General Background

Modern protein research is operationally fragmented. A researcher studying a disease-linked protein must navigate at minimum four to six separate platforms to complete a single research task: UniProt for sequence and annotation data, AlphaFold DB or SWISS-MODEL for structural prediction, UCSF Chimera or PyMOL for 3D visualization, STRING or BioGRID for interaction data, and DrugBank for pharmacological context. None of these platforms interoperate. Data must be manually transferred between them. Visualization tools require OS-specific desktop installations. The result is that significant researcher time is spent on infrastructure navigation rather than on the biological science itself.

Protein Language Models (PLMs) — particularly ESM-2, developed by Meta AI — now provide high-quality learned representations of protein sequences that enable machine learning models to predict protein function, interaction, and mutational effects with accuracy approaching experimental benchmarks. These models have not yet been integrated into accessible, unified research platforms accessible to the broader scientific community without computational expertise.

ProtMind addresses both problems simultaneously: it unifies the fragmented tool landscape into one browser-based platform, and it integrates state-of-the-art PLM-based prediction models with full explainability and evidence tracing — accessible to researchers and clinicians without command-line expertise.

1.5 Research Problem

Core Research Problem

How can an integrated, AI-powered platform unify protein sequence analysis, structural prediction, interaction modeling, and clinical evidence synthesis into a single accessible system that reduces researcher workflow friction, provides explainable and evidence-backed predictions, and produces clinically actionable outputs for rare diseases such as Cholangiocarcinoma?

The research problem decomposes into four sub-problems:

1. Infrastructure Fragmentation: Bioinformatics workflows require 4–6 separate platforms with no interoperability, imposing significant overhead on every research task.
2. OS and Access Barriers: State-of-the-art 3D visualization tools require desktop installation on specific OS configurations, excluding researchers without those configurations.
3. Black-Box AI Predictions: Existing AI-based protein function and interaction predictors return scores without biological explanation, limiting their utility and trustworthiness for clinical applications.
4. Rare Disease Research Gaps: Diseases like Cholangiocarcinoma (CCA) are understudied at the protein level despite available genomic data, creating a gap that better tooling could begin to close.

1.6 Product Scope

ProtMind is a web-based unified bioinformatics research platform. Its scope encompasses:

In Scope	Out of Scope
Browser-based 3D protein structure visualization	Native desktop application development
AI-powered protein function prediction (GO MF terms)	Protein structure determination from raw experimental data (NMR/X-ray)
Protein-protein interaction prediction and visualization	Drug discovery pipeline execution (wet-lab integration)
Cross-species essentiality and mutation effect prediction	Electronic Health Record (EHR) integration
CCA biomarker and drug target candidate identification	Clinical trial management or patient-facing medical tools
Evidence-backed explainability for all predictions	Real-time genomic sequencing data ingestion
Unified dashboard with natural language and direct tool access	Full-genome variant calling pipelines
Data sourced from UniProt, GO, PDB, AlphaFold, BioGRID, STRING, DrugBank, TCGA	Proprietary or non-public database integration

1.7 Project Objectives

5. Build a clean, version-controlled, reproducible biological dataset from six major public databases with full data cards and pipeline documentation.
6. Deliver six standalone browser-based visualization tools that replace the current multi-platform desktop workflow for protein structure exploration.
7. Train and deploy a protein function prediction model (Model 1) using ESM-2 embeddings with residue-level explainability, evaluated on Fmax, AUPR, and MCC.
8. Implement a full trust and evidence layer that makes every prediction verifiable, calibrated, and traceable to source databases and literature.
9. Train and deploy a protein-protein interaction prediction model (Model 2) with structural binding interface visualization and pharmacological context.
10. Train and deploy a cross-species essentiality and mutation effect prediction model (Model 3) with structural comparison and RMSD quantification.

11. Produce a complete end-to-end clinical demonstration using CCA TCGA data — from raw expression profiles to ranked, evidence-backed biomarker and drug target candidates.
12. Deliver a fully documented, reproducible codebase suitable for academic publication and continued development.

1.8 References

Reference	Source
UniProt Knowledgebase (Swiss-Prot)	https://www.uniprot.org — reviewed protein entries only
Gene Ontology (GO) and GOA	https://geneontology.org — experimental evidence codes only
Protein Data Bank (PDB)	https://www.rcsb.org — experimentally determined structures
AlphaFold Protein Structure Database	https://alphafold.ebi.ac.uk — AI-predicted structures
SWISS-MODEL Homology Modeling	https://swissmodel.expasy.org
BioGRID Interaction Database	https://thebiogrid.org
STRING Interaction Network	https://string-db.org
DrugBank — Drug and Target Database	https://www.drugbank.com
TCGA via NCI GDC	https://portal.gdc.cancer.gov — CCA expression data
ESM-2 Protein Language Model	Lin et al., 2023. Meta AI Research
Mol* 3D Viewer	https://molstar.org — WebGL-based browser protein viewer
3Dmol.js	https://3dmol.csb.pitt.edu — alternative WebGL viewer

Chapter 2: Overall Description

2.1 Product Perspective

ProtMind is a new, self-contained system that does not replace any single existing product but replaces the workflow that currently requires several products used in combination. It sits in the intersection of three product categories: bioinformatics portals (UniProt, NCBI), structure visualization tools (UCSF Chimera, PyMOL), and AI prediction platforms (AlphaFold, PredictProtein).

> **System Context**
> ProtMind is positioned as a research decision-support platform, not a standalone automated pipeline. It provides the researcher with enhanced tools, AI-generated predictions, and unified evidence — but the researcher remains the scientific decision-maker. Every output is designed to support judgment, not to replace it.

The system interacts with the following external systems through APIs and data ingestion pipelines:

- AlphaFold DB — queried for existing predicted structures via REST API
- SWISS-MODEL — called for homology modeling for sequences not covered by AlphaFold
- UniProt REST API — protein sequence and annotation retrieval
- PDB REST API — experimental structure retrieval
- STRING API — protein interaction network data
- DrugBank — pharmacological compound and target data
- TCGA / NCI GDC — CCA expression and genomic data (ingested into local database)

2.2 Product Functions

ProtMind's top-level functions are organized by the five system layers:

Function Category	Key Capabilities
Data Management	Ingest, normalize, version, and serve data from 6 public databases; time-based train/test splitting
Visualization Suite	6 standalone browser-based tools: structure builder, 3D viewer, sequence annotator, structure comparator, interaction network, expression heatmap
Protein Function Prediction	ESM-2 + MLP/XGBoost; GO MF term prediction with confidence scores and residue-level explainability

PPI Prediction	Paired ESM-2 embeddings; interaction type, residues, binding interface, pharmacological relevance
Essentiality & Mutation Analysis	Cross-species PLM; essentiality scores, mutation impact (neutral/mildly/severely disruptive), RMSD comparison
Explainability & Trust	Evidence Graph, calibration curves, attention heatmaps, citation-ready references for every prediction
CCA Clinical Demonstration	End-to-end TCGA pipeline: differential expression → candidate ranking → biomarker/drug target identification
Research Dashboard	Unified web interface: natural language queries + direct tool access on any OS, any browser

2.3 User Classes and Characteristics

User Class	Description	Technical Level	Primary Use
Biomedical Researcher	Academic or industry researcher studying protein biology	High — domain expert, moderate computational	Protein function queries, structure exploration, literature synthesis
Computational Biologist	Researcher with strong computational and bioinformatics background	Very high — comfortable with APIs, pipelines	Model outputs, data exports, pipeline integration
Clinician / Medical Professional	Doctor or clinical scientist evaluating protein-level disease mechanisms	Low-moderate — domain expert, minimal computation	Evidence graphs, CCA candidate lists, structured reports
Graduate Student	Masters or PhD student learning protein research methods	Moderate — learning bioinformatics tools	Visualization tools, function prediction, tutorial workflows
Platform Administrator	System operator managing infrastructure, data updates, and monitoring	Technical — DevOps/SRE background	System health, data pipeline management, access control

2.4 Operating Environment

Component	Specification
Client — Browser	Any modern browser (Chrome 90+, Firefox 85+, Safari 14+, Edge 90+) with WebGL support enabled
Client — OS	Operating system independent — macOS, Windows, Linux, ChromeOS all supported
Client — Hardware	Minimum: 4GB RAM, integrated GPU with WebGL. Recommended: 8GB RAM, dedicated GPU for smooth 3D rendering
Server — Compute	Cloud-hosted (initially single-node, scalable to multi-node after MVP compute access secured)
Server — OS	Ubuntu 22.04 LTS
Server — GPU	NVIDIA A100 or equivalent (required for Phase 3+ ESM-2 embedding extraction and model training)
Network	Minimum 10 Mbps for 3D rendering; 100 Mbps recommended for large structure file transfers
Database	PostgreSQL 15+ (relational), with object storage (S3-compatible) for PDB/structure files
Containerization	Docker + Docker Compose for development; Kubernetes for production scaling

2.5 Design and Implementation Constraints

- Data Licensing: All data sources (UniProt, GO, PDB, AlphaFold, BioGRID, STRING, DrugBank, TCGA) must be used in compliance with their respective open-access licenses. Commercial usage restrictions must be reviewed before any commercialization.
- Compute Dependency: ESM-2 embedding extraction and model training require GPU resources not available in Phase 1–2. Phases 3–6 are gated on securing compute access through the MVP demonstration.
- Browser WebGL Requirement: The 3D visualization layer requires WebGL support. Fallback behavior for WebGL-disabled environments must be defined (static image rendering fallback).
- Experimental Evidence Only: GO annotation labels used for training must be filtered to experimental evidence codes (EXP, IDA, IPI, IMP, IGI, IEP). Computationally predicted annotations must be excluded to prevent label leakage.
- Time-Based Data Splits: Dataset splitting must be time-based (by annotation date), not random, to prevent temporal data leakage and ensure the model's performance reflects genuine predictive value on future annotations.
- Single-Platform Constraint: The system is designed as a web application with no OS-specific components. Desktop application development is out of scope.
- AlphaFold DB Coverage Limit: AlphaFold DB does not cover all protein sequences. The SWISS-MODEL fallback must handle sequences not in AlphaFold DB gracefully, with latency expectations set accordingly.

2.6 Stakeholder Analysis

Stakeholder	Role	Primary Interest	Influence Level
Biomedical Research Community	Primary users	Faster, trustworthy protein research tools	High — adoption defines success
Clinical Scientists (CCA Focus)	Validation audience	Actionable biomarker and drug target candidates	High — clinical credibility
Academic Institutions	Affiliates / collaborators	Publication-quality outputs, reproducibility	Medium
Funding Bodies / Grant Committees	Resource providers	MVP demonstration quality, scientific rigor	High — gate compute access
Data Providers (UniProt, PDB, etc.)	Data sources	Correct attribution, license compliance	Medium
Open-Source Community (Mol*, 3Dmol.js)	Tool contributors	Correct integration and attribution	Low-Medium
Platform Development Team	Builders	Technical feasibility, clear requirements	Very High

2.7 Non-Functional Requirements (NFR) — Summary

NFR Category	High-Level Requirement
Performance	3D structure rendering must complete within 3 seconds for structures up to 500 residues on recommended hardware
Scalability	System must support at least 100 concurrent users without degradation during MVP phase
Reliability	Dashboard uptime target: 99.5% during active research sessions; data pipeline runs must be idempotent
Security	All user queries and protein data transmitted over HTTPS/TLS 1.3; no personal health information stored
Usability	A researcher unfamiliar with the platform must be able to obtain a protein function prediction within 5 minutes of first use
Reproducibility	All model training runs, data pipeline executions, and outputs must be fully reproducible from documented artifacts

Portability	All visualization tools must render identically on macOS, Windows, and Linux in supported browsers
Explainability	Every model prediction must be accompanied by residue-level evidence, confidence score, and at least one supporting database reference

2.8 Feasibility Study

Technical Feasibility

ProtMind's technical approach builds on well-established open-source foundations. ESM-2 is publicly available and has been validated in peer-reviewed literature. Mol* and 3Dmol.js are production-grade WebGL viewers already in use on major bioinformatics portals. UniProt, GO, PDB, AlphaFold, BioGRID, STRING, DrugBank, and TCGA all provide free API access. The primary technical risk is GPU compute availability, which is managed by gating the heavier model phases behind the MVP demonstration.

Operational Feasibility

The platform is designed for the existing workflows of biomedical researchers and clinicians. The dashboard interface requires no programming knowledge for basic operation. The visualization tools replace existing workflows that researchers already perform manually — the learning curve is the elimination of steps, not the addition of them. Researcher adoption is supported by the standalone visualization tools (Phase 2 value before AI is deployed) and by structured, citation-ready outputs that fit into academic workflows.

Economic Feasibility

Phase 1 and Phase 2 can be executed using standard cloud compute (CPU-only, low cost). Phase 3 onward requires GPU access, which is the target of the MVP demonstration to grant committees or institutional HPC resources. All data sources are freely available for research use. Development tooling (FastAPI, React, PostgreSQL, PyTorch) is open-source and incurs no licensing cost.

2.9 System Assumptions and Dependencies

Assumption / Dependency	Rationale
UniProt Swiss-Prot reviewed entries remain publicly accessible	Entire data pipeline depends on Swiss-Prot annotations
AlphaFold DB REST API remains available and stable	Structure builder's primary data source
SWISS-MODEL API accepts programmatic structure requests	Fallback for sequences not in AlphaFold DB

ESM-2 model weights remain publicly available (Meta AI)	Core of all three prediction models
WebGL is supported by the user's browser	Required for all 3D visualization tools
TCGA CCA expression data is available via NCI GDC portal	Required for Phase 7 clinical demonstration
GPU compute is accessible after Phase 3 MVP demonstration	Required for Phases 3–6 model training
GO experimental evidence codes correctly distinguish validated from predicted annotations	Critical for training data quality
Internet connectivity is available during API-dependent operations	Required for real-time AlphaFold DB and SWISS-MODEL queries

Chapter 3: External Interfaces & System Analysis

3.1 User Interfaces

ProtMind's user interface is organized into a unified web dashboard accessible through any modern browser. The interface supports two primary interaction modes: natural language query input (AI-agent mode) and direct tool access (standalone mode).

3.1.1 Main Interface Views

View	Description	Access Mode
Landing Dashboard	Entry point: search bar for natural language queries, quick-access tiles for each visualization tool	All users
Protein Search & Overview	Enter UniProt ID or sequence — shows basic protein metadata, organism, disease associations	All users
Sequence-to-Structure Builder	Input UniProt ID or raw sequence → renders 3D model with pLDDT coloring and PAE plot	Standalone + AI-integrated
3D Structure Viewer	Upload PDB / enter PDB ID or AlphaFold accession → full interactive 3D rendering	Standalone + AI-integrated
Sequence Annotator	Displays annotated amino acid sequence with GO regions, PTMs, active sites, ESM-2 heatmap overlay	Standalone + AI-integrated
Structure Comparator	Input two structures → overlaid 3D view with RMSD score and difference highlighting	Standalone + AI-integrated
Interaction Network Viewer	Interactive PPI graph — expandable nodes, filterable edges, click-through to pharmacological context	Standalone + AI-integrated
Expression Heatmap	TCGA expression data across tissue types and disease conditions	Standalone + AI-integrated
Function Prediction Report	AI-generated GO MF prediction with confidence scores, residue heatmap, and evidence graph	AI-integrated only
PPI Prediction Report	Interaction prediction with binding interface visualization, network context, drug relevance	AI-integrated only
Mutation / Essentiality Report	Essentiality scores, mutation impact, WT vs. mutant RMSD overlay, clinical flags	AI-integrated only

CCA Biomarker Dashboard	Ranked candidate list from TCGA differential expression → full pipeline evidence per candidate	AI-integrated only

3.2 Hardware Interfaces

3.2.1 Server-Side Requirements

Component	Minimum Specification	Recommended Specification
CPU	8-core x86-64 (Phase 1–2)	32-core x86-64 (Phase 3+)
RAM	16 GB (Phase 1–2)	128 GB (Phase 3+ model serving)
GPU	None required (Phase 1–2)	NVIDIA A100 80GB or 4× A40 48GB (Phase 3+)
Storage	500 GB SSD (data + pipeline)	4 TB NVMe SSD (full dataset + model checkpoints)
Network	1 Gbps	10 Gbps (high-throughput structure serving)
Object Storage	S3-compatible bucket for PDB/structure files	AWS S3 or equivalent with CDN

3.2.2 Client-Side Requirements

Requirement	Minimum	Recommended
RAM	4 GB	8 GB+
GPU	Integrated GPU (WebGL 1.0 support)	Dedicated GPU (WebGL 2.0 for high-quality 3D)
Screen Resolution	1280×720	1920×1080 or higher
Browser	Chrome 90+, Firefox 85+, Safari 14+, Edge 90+	Chrome 110+ or Firefox 100+
Internet Connection	10 Mbps	50 Mbps+ (smooth large structure loading)

3.3 Software Interfaces

ProtMind integrates with multiple external software systems via REST APIs and local data ingestion pipelines. All external calls are handled through a dedicated API Gateway layer to enforce rate limiting, error handling, and caching.

3.3.1 External API Integrations

External System	Interface Type	Data Retrieved	Phase
UniProt REST API	REST/JSON	Protein sequences, organism, feature annotations, disease associations	Phase 1+
AlphaFold DB API	REST/JSON + mmCIF/PDB file download	Predicted 3D structure, pLDDT scores, PAE matrix	Phase 2+
SWISS-MODEL API	REST/JSON + PDB file download	Homology models for sequences not in AlphaFold DB	Phase 2+
PDB REST API	REST/JSON + PDB file download	Experimentally determined 3D structures	Phase 2+
STRING API v11.5	REST/JSON	Protein interaction network data with confidence scores	Phase 1+
BioGRID API	REST/JSON	Curated protein-protein interaction data	Phase 1+
DrugBank API	REST/JSON	Drug-target associations, pharmacological data	Phase 5+
TCGA / NCI GDC API	REST/JSON	CCA expression data, genomic annotations	Phase 1 (ingestion)

3.4 Communications Interfaces

- HTTPS/TLS 1.3: All client-server communication encrypted in transit. No unencrypted HTTP endpoints in production.
- WebSocket: Used for real-time progress updates during long-running operations (structure building, model inference, homology modeling).
- REST API (Internal): Frontend communicates with backend via versioned REST API (v1). All responses follow a consistent JSON envelope structure.
- Server-Sent Events (SSE): Used as fallback for WebSocket in environments where persistent connections are restricted.
- Rate Limiting: All outbound calls to external APIs (AlphaFold, SWISS-MODEL, STRING, etc.) are rate-limited and cached to prevent abuse and comply with provider terms.
- CORS Policy: Configured to allow only trusted origins in production; permissive in development environments.

3.5 System Analysis

The system operates in two primary interaction flows:

Flow A — AI-Agent Driven Query

13. User submits a natural language query (e.g., 'What is the function of P01308 and does it interact with insulin receptor?')
14. LLM Router parses intent and identifies required tools/models
15. Router orchestrates sequential or parallel calls: data retrieval → embedding extraction → model inference → visualization rendering
16. Each step's output is passed to the Output Formatter
17. User receives a structured response: plain-language summary, step-by-step explanation, technical details, visualizations, and references

Flow B — Direct Standalone Tool Access

18. User opens a visualization tool directly from the dashboard (e.g., '3D Structure Viewer')
19. User enters a UniProt ID, PDB ID, or uploads a file
20. Tool fetches required data from the appropriate API or local database
21. Visualization renders immediately in the browser with full interactive controls
22. No AI agent involvement — pure tool-use mode

3.6 Stakeholder Analysis (Detailed)

3.6.1 Biomedical Researcher

Attribute	Description
Goals	Efficiently query protein function, structure, and interactions; export results for publication
Pain Points	Current multi-tab workflow; OS-restricted desktop tools; no unified evidence view
Success Criteria	Complete a protein research task in one platform session without external tool switching
Key Features	All visualization tools, function prediction, PPI prediction, evidence graph, report export

3.6.2 Clinical Scientist / CCA Researcher

Attribute	Description
Goals	Identify protein-level biomarkers and drug targets relevant to CCA; understand evidence behind candidates

Pain Points	No integrated platform connecting genomics → protein function → clinical context
Success Criteria	Receive a ranked list of CCA biomarker candidates with full supporting evidence, exportable as a structured report
Key Features	CCA biomarker dashboard, expression heatmap, full pipeline reports, PDF export

3.6.3 Graduate Student / Trainee Researcher

Attribute	Description
Goals	Learn protein research workflows; explore protein structures and functions for thesis/coursework
Pain Points	Steep learning curve of current bioinformatics tools; OS barriers; no guided workflow
Success Criteria	Explore a protein from sequence to structure to function without external documentation
Key Features	Sequence-to-structure builder, sequence annotator, guided dashboard, natural language query

3.6.4 Platform Administrator

Attribute	Description
Goals	Keep the system running, data current, and pipelines healthy; manage access and monitor usage
Pain Points	Manual data pipeline maintenance; no visibility into system health without custom tooling
Success Criteria	Re-run data pipeline on new UniProt release with zero manual data transformation; monitor error rates via dashboard
Key Features	Pipeline management, data versioning, system monitoring, access control, audit logs

3.7 Functional Requirements (FR)

3.7.1 Data Layer & Pipeline

FR ID	Requirement	Phase	Priority

FR-001	System SHALL ingest Swiss-Prot reviewed UniProt entries via REST API	Phase 1	Critical
FR-002	System SHALL filter GO annotations to experimental evidence codes only (EXP, IDA, IPI, IMP, IGI, IEP)	Phase 1	Critical
FR-003	System SHALL implement time-based dataset splitting using annotation date fields	Phase 1	Critical
FR-004	System SHALL version all dataset artifacts with data cards documenting source, date, and processing steps	Phase 1	High
FR-005	System SHALL ingest PDB and AlphaFold DB structure files into object storage	Phase 1	High
FR-006	System SHALL ingest BioGRID and STRING interaction data into the Interactions table	Phase 1	High
FR-007	System SHALL ingest TCGA CCA expression data and normalize it for the Expression Heatmap	Phase 1	High
FR-008	Pipeline SHALL be re-runnable on new UniProt releases without manual intervention	Phase 1	High

3.7.2 Visualization Suite

FR ID	Requirement	Phase	Priority
FR-009	System SHALL accept a UniProt ID or raw amino acid sequence and return a 3D structural model	Phase 2	Critical
FR-010	System SHALL query AlphaFold DB first; fall back to SWISS-MODEL if structure not found	Phase 2	Critical
FR-011	System SHALL render 3D protein structures in the browser using WebGL (Mol* or 3Dmol.js)	Phase 2	Critical
FR-012	3D viewer SHALL support cartoon, surface, and ball-and-stick representation modes	Phase 2	High
FR-013	3D viewer SHALL color structures by pLDDT confidence score with blue/orange/red scale	Phase 2	High
FR-014	System SHALL display PAE (Predicted Aligned Error) plot alongside predicted structures	Phase 2	High
FR-015	Sequence Annotator SHALL display GO-relevant regions, active sites, PTM sites, signal peptides, and transmembrane domains	Phase 2	High

FR-016	Structure Comparator SHALL accept two structures and compute RMSD with overlaid rendering	Phase 2	High
FR-017	Interaction Network Viewer SHALL render PPI networks as interactive expandable graphs	Phase 2	High
FR-018	Expression Heatmap SHALL visualize TCGA expression data across tissue types	Phase 2	High
FR-019	All visualization tools SHALL be accessible as standalone features without AI query involvement	Phase 2	Critical
FR-020	3D viewer SHALL allow export of rendered images in PNG format	Phase 2	Medium

3.7.3 Model 1 — Protein Function Prediction

FR ID	Requirement	Phase	Priority
FR-021	System SHALL extract ESM-2 embeddings for all proteins in the pilot dataset	Phase 3	Critical
FR-022	System SHALL train a multi-label classifier for GO Molecular Function prediction	Phase 3	Critical
FR-023	System SHALL evaluate Model 1 using Fmax, AUPR, and MCC on the time-based test set	Phase 3	Critical
FR-024	Function prediction output SHALL include per-GO-term confidence scores	Phase 3	Critical
FR-025	System SHALL generate residue-level attention heatmaps using ESM-2 attention weights	Phase 3	High
FR-026	Function prediction output SHALL be linked to the Sequence Annotator view	Phase 3	High
FR-027	Function prediction output SHALL be linked to the 3D Viewer with highlighted residue regions	Phase 3	High

3.7.4 Explainability & Trust Layer

FR ID	Requirement	Phase	Priority
FR-028	System SHALL generate an Evidence Graph for every prediction showing sequence, structural, and literature evidence	Phase 4	Critical

FR-029	System SHALL implement and report confidence calibration curves for all models	Phase 4	High
FR-030	System SHALL apply Platt or temperature scaling to correct miscalibrated predictions	Phase 4	High
FR-031	Mutation analysis SHALL identify whether a mutation falls in an active site, allosteric site, or drug-binding region	Phase 4	High
FR-032	Every prediction output SHALL include at least one verifiable database reference	Phase 4	Critical

3.7.5 Model 2 — Protein-Protein Interaction Prediction

FR ID	Requirement	Phase	Priority
FR-033	System SHALL accept two protein inputs and predict interaction probability and type	Phase 5	Critical
FR-034	PPI prediction SHALL use paired ESM-2 embeddings (concatenation + difference + product)	Phase 5	Critical
FR-035	PPI output SHALL include predicted interaction residues and binding interface visualization	Phase 5	High
FR-036	PPI output SHALL query DrugBank for compounds targeting the interaction	Phase 5	High
FR-037	PPI output SHALL show the protein pair in their broader interaction network context	Phase 5	High

3.7.6 Model 3 — Cross-Species Essentiality & Mutation Analysis

FR ID	Requirement	Phase	Priority
FR-038	System SHALL predict protein essentiality scores across human, mouse, and yeast	Phase 6	Critical
FR-039	System SHALL predict mutation effect as neutral, mildly disruptive, or severely damaging	Phase 6	Critical
FR-040	Mutation analysis SHALL automatically render WT vs. mutant Structure Comparator overlay	Phase 6	High
FR-041	System SHALL support batch analysis of TCGA differentially expressed proteins for network-level essentiality mapping	Phase 6	High

3.7.7 CCA Clinical Demonstration

FR ID	Requirement	Phase	Priority
FR-042	System SHALL identify significantly differentially expressed proteins in CCA tumor vs. normal tissue using TCGA data	Phase 7	Critical
FR-043	System SHALL run CCA candidate proteins through the full ProtMind pipeline automatically	Phase 7	Critical
FR-044	System SHALL produce a ranked list of CCA biomarker and drug target candidates with full evidence	Phase 7	Critical
FR-045	System SHALL export the full evidence graph and predictions as a structured PDF report	Phase 7	High

3.8 Non-Functional Requirements (NFR)

3.8.1 Security

- NFR-001: All data in transit MUST be encrypted using TLS 1.3 or higher.
- NFR-002: No personally identifiable information (PII) or protected health information (PHI) SHALL be stored on the platform.
- NFR-003: API keys for external services MUST be stored in environment variables, never hardcoded in source code.
- NFR-004: Authentication tokens MUST expire and support refresh token rotation.
- NFR-005: SQL injection, XSS, and CSRF protections MUST be implemented across all API endpoints.

3.8.2 Performance

- NFR-006: 3D structure rendering for proteins up to 500 residues MUST complete within 3 seconds on recommended client hardware.
- NFR-007: Protein function prediction inference MUST complete within 10 seconds for a single protein.
- NFR-008: Dashboard initial load time MUST be under 2 seconds on a 50 Mbps connection.
- NFR-009: API responses for metadata queries (not structure files) MUST be under 500ms at the 95th percentile.

3.8.3 Scalability

- NFR-010: The system MUST support at least 100 concurrent users during MVP phase without performance degradation.
- NFR-011: The data pipeline MUST process a full UniProt Swiss-Prot refresh within 24 hours.
- NFR-012: The architecture MUST support horizontal scaling of the API layer without application changes.

3.8.4 Usability & UX

- NFR-013: A researcher new to the platform MUST be able to complete a protein function prediction within 5 minutes of first use without documentation.
- NFR-014: All visualization tools MUST be accessible within 2 clicks from the main dashboard.
- NFR-015: Error messages MUST be descriptive and suggest corrective action (not raw stack traces).
- NFR-016: The interface MUST be responsive and usable on screens from 1280×720 to 4K resolution.

3.8.5 Reproducibility & Scientific Integrity

- NFR-017: All model training runs MUST be fully reproducible from a documented configuration file (random seeds, data versions, hyperparameters).
- NFR-018: Every prediction output MUST include the model version and dataset version used to generate it.
- NFR-019: The data pipeline MUST be idempotent — running it twice on the same inputs produces identical outputs.

3.8.6 Portability & Reliability

- NFR-020: The visualization layer MUST render identically on macOS, Windows, and Linux in all supported browsers.
- NFR-021: Dashboard uptime MUST be 99.5% or higher during active research periods.
- NFR-022: All external API calls MUST implement retry logic with exponential backoff and graceful degradation.
- NFR-023: The system MUST provide fallback behavior (static rendering) when WebGL is unavailable.

3.9 Feasibility Study

3.9.1 Technical Feasibility

Technical Component	Feasibility Assessment	Risk Level
ESM-2 embeddings extraction	High — publicly available model with documented API	Low
WebGL 3D visualization (Mol*, 3Dmol.js)	High — production-ready libraries in use on major portals	Low
AlphaFold DB API integration	High — stable REST API with documented endpoints	Low

SWISS-MODEL homology modeling (fallback)	Medium — depends on SWISS-MODEL API availability and rate limits	Medium
Multi-label GO MF classification	High — well-studied problem with established baselines	Low
GPU compute for training	Medium — available after MVP; gated on resource acquisition	Medium
Cross-species essentiality model (Phase 6)	Medium-Low — less established; depends on training data quality	High
PDF report export from evidence graph	High — standard tooling (Puppeteer, WeasyPrint)	Low

3.9.2 Operational Feasibility

The platform directly addresses workflows that researchers already perform manually. The value proposition is reduction of steps, not addition of new processes. Early phases (1–2) deliver standalone visualization value before any AI model is deployed, reducing the risk that the platform needs to demonstrate prediction quality before earning user trust. The structured output format (summary → steps → details → references) mirrors how research findings are already communicated in the field, lowering the cognitive barrier to adoption.

3.9.3 Economic Feasibility

Phase	Compute Cost	Data Cost	Notes
Phase 1 — Data Pipeline	Low (CPU only)	Free	All sources open-access; standard cloud VM
Phase 2 — Visualization Suite	Low (CPU only)	Free	WebGL rendering is client-side
Phase 3 — MVP Model 1	Medium (GPU required)	Free	A100 or equivalent for ESM-2 extraction
Phase 4 — Explainability	Low-Medium	Free	Calibration and evidence layer are compute-light
Phase 5 — Model 2	Medium (GPU required)	Free	Paired embedding computation is more expensive
Phase 6 — Model 3	High (GPU required)	Free	Cross-species model is most compute-intensive
Phase 7 — CCA Demo	Medium	Free	Pipeline runs on already-trained models

3.10 Use Case Mapping

Use Case	Actor	Related FR	Phase
UC-001: Query protein function by UniProt ID	Researcher	FR-021 to FR-027	Phase 3
UC-002: Explore 3D structure interactively	Researcher	FR-009 to FR-014	Phase 2
UC-003: Compare wild-type vs. mutant structure	Researcher	FR-016, FR-040	Phase 2, 6
UC-004: View annotated protein sequence	Researcher	FR-015, FR-025	Phase 2, 3
UC-005: Predict protein-protein interaction	Researcher	FR-033 to FR-037	Phase 5
UC-006: Explore interaction network	Researcher	FR-017, FR-037	Phase 2, 5
UC-007: Analyze mutation effect	Researcher	FR-031, FR-039, FR-040	Phase 4, 6
UC-008: View CCA expression heatmap	Clinical Scientist	FR-018, FR-007	Phase 2, 7
UC-009: Generate CCA	Clinical Scientist	FR-042 to FR-045	Phase 7

biomarker candidate report			
UC-010: Export prediction as PDF report	Researcher	FR-045	Phase 7
UC-011: Re-run data pipeline on new UniProt release	Platform Admin	FR-008	Phase 1
UC-012: Monitor system health and pipeline status	Platform Admin	FR-004	Phase 1

3.11 System Constraints (Detailed Analysis)

3.11.1 Schedule and Time Constraints

The project is structured in seven sequential phases with dependencies between them. Phases 1–2 have no GPU compute dependency and can proceed immediately. Phase 3 is the critical milestone that gates GPU resource allocation. The sequencing of visualization tools before models (Phase 2 before Phase 3) is intentional — it delivers demonstrable value early and tests infrastructure before model integration begins.

- Phase 1–2 are CPU-only and can begin immediately with standard cloud resources.
- Phase 3 is gated on GPU availability — delays in securing compute directly delay Phases 3–6.
- Phase 7 depends on all three models being trained and evaluated, making it the terminal phase.

3.11.2 Financial and Resource Constraints

- All data sources are free for research use; however, commercial use of DrugBank data requires licensing.
- GPU compute (Phase 3+) requires either institutional HPC access, cloud GPU instances, or research grant allocation.
- The system is designed to minimize infrastructure cost during Phases 1–2 — standard web hosting is sufficient.

3.11.3 Legal and Compliance Constraints

- All six data sources must be used in compliance with their respective licenses. No patient-level data from TCGA may be stored or exposed beyond what is permitted by the TCGA data use agreement.
- ESM-2 model weights are released under the MIT license — commercial use is permitted with attribution.
- Mol* and 3Dmol.js are released under open-source licenses compatible with research use.

3.11.4 Technical and Environmental Constraints

- WebGL is a hard dependency for 3D visualization. Environments that block WebGL (some corporate browsers, privacy extensions) will experience degraded functionality.
- SWISS-MODEL API rate limits constrain how many homology modeling requests can be made per session; caching and queuing are required.
- AlphaFold DB covers approximately 200 million protein sequences but does not cover all possible sequences — especially novel variants.

3.11.5 Data Scope and Security Constraints

- The platform handles protein research data only. No patient data, no clinical records, and no identifiable personal information is in scope.
- TCGA expression data used is aggregate/summary-level, not individual patient records.
- All database credentials and API keys must be managed through a secrets manager in production.

3.12 System Assumptions (Operational Premises)

3.12.1 Infrastructure and Connectivity Assumptions

- Researchers have reliable internet connectivity during active platform sessions (required for real-time AlphaFold DB and SWISS-MODEL queries).
- The hosting environment provides persistent object storage for PDB and structure files.
- SSL/TLS certificates are available and automatically renewed in the production environment.

3.12.2 Data Quality Assumptions

- UniProt Swiss-Prot reviewed entries are assumed to be of higher quality than TrEMBL unreviewed entries for training purposes.
- GO experimental evidence codes accurately distinguish experimentally validated from computationally predicted annotations.
- AlphaFold DB structure predictions are of sufficient quality for visualization and as training features, with pLDDT and PAE scores as quality proxies.

3.12.3 Scientific and Domain Assumptions

- ESM-2 embeddings capture sufficient protein sequence information to enable accurate function, interaction, and essentiality predictions.
- TCGA CCA expression data is representative of CCA biology at the protein level and appropriate for biomarker candidate identification.
- Time-based data splitting (using annotation dates) is a scientifically valid approach to preventing temporal leakage in protein function prediction benchmarks.

3.12.4 User Behavior Assumptions

- Users are assumed to have a basic understanding of protein biology concepts (sequences, structures, GO terms). The system provides context but is not designed as a primary biology education tool.
- Users are assumed to operate in good faith — the system is a research tool, not a public consumer application.
- Platform administrators are assumed to have basic Linux and Docker proficiency for system operations.

Chapter 4: System Design

4.1 System Architecture (High-Level Design)

4.1.1 Architectural Style

ProtMind follows a layered service-oriented architecture with a clear separation between the frontend presentation layer, backend API layer, ML inference services, and data layer. The architecture prioritizes modularity — each visualization tool and each prediction model is a deployable service unit — enabling independent scaling and phased delivery.

4.1.2 High-Level Architecture Overview

```
┌──────────────────────────────────────────────────────────┐
│              PRESENTATION LAYER (React Frontend)          │
│  Dashboard UI  |  Viz Tools  |  Report Views              │
└──────────────────────────────────────────────────────────┘
                              │
                              │ REST API / WebSocket
┌──────────────────────────────────────────────────────────┐
│           API GATEWAY + LLM ROUTER (FastAPI)              │
│  Auth  |  Rate Limiting  |  Task Orchestration            │
└──────────────────────────────────────────────────────────┘
   │          │              │                        │
   ▼          ▼              ▼                        ▼
[Data Svc]  [Viz Engine]  [ML Inference]  [External APIs]
   │          │              │                        │
┌──────────────────────────────────────────────────────────┐
│                        DATA LAYER                         │
│  PostgreSQL  |  Object Storage  |  Embedding Cache        │
└──────────────────────────────────────────────────────────┘
```

4.1.3 Major System Components

Component	Technology	Responsibility
Frontend (SPA)	React + TypeScript	Dashboard UI, visualization rendering, user interaction
3D Visualization Engine	Mol* / 3Dmol.js (WebGL)	Browser-based 3D structure rendering
API Gateway	FastAPI (Python)	Request routing, authentication, rate limiting, WebSocket
LLM Router / Orchestrator	LangChain + OpenAI/local LLM	Natural language query parsing, tool selection, pipeline orchestration
Data Service	FastAPI + SQLAlchemy	Protein metadata, annotation, and interaction queries
ML Inference Service	PyTorch + ESM-2	Function, PPI, and essentiality model inference

Data Pipeline	Apache Airflow / Prefect	Scheduled ingestion and normalization from external DBs
Primary Database	PostgreSQL 15	Relational data: proteins, annotations, interactions, dataset splits
Object Storage	S3-compatible	PDB files, structure files, model checkpoints
Embedding Cache	Redis	Cached ESM-2 embeddings to avoid re-computation
Task Queue	Celery + Redis	Async jobs: structure building, model inference, report generation

4.1.4 API Communication Model

- All frontend-to-backend communication uses versioned REST endpoints under /api/v1/.
- Long-running operations (homology modeling, model inference) return a job ID immediately and push progress via WebSocket.
- The frontend polls /api/v1/jobs/{id}/status for environments that cannot maintain WebSocket connections.
- External API calls (AlphaFold, SWISS-MODEL, STRING) are made by the backend only — never directly from the frontend — to protect API keys and enable caching.

4.1.5 Security Architecture

- JWT-based authentication with refresh token rotation. Access tokens expire in 15 minutes; refresh tokens in 7 days.
- Role-based access control (RBAC): Researcher, Clinical Scientist, Platform Admin roles with scoped permissions.
- All external API keys stored in environment variables via a secrets manager (AWS Secrets Manager or equivalent).
- Database credentials rotated automatically via infrastructure-as-code tooling.

4.1.6 Logical Deployment View

Environment	Description	Services Running
Development	Local Docker Compose	All services on single machine; mock external APIs available
Staging	Cloud (CPU only)	Full stack without GPU inference; static ESM-2 embeddings for testing
Production (Phase 1–2)	Cloud VM (CPU)	Frontend, API, Data Service, PostgreSQL, Object Storage
Production (Phase 3+)	Cloud VM + GPU node	All Phase 1–2 services + ML Inference Service on GPU instance

4.2 Technology Stack

4.2.1 Backend Technologies

Technology	Version	Purpose
Python	3.11+	Primary backend language
FastAPI	0.110+	REST API framework — async, type-safe, OpenAPI auto-docs
PyTorch	2.0+	ML model training, inference, ESM-2 integration
ESM-2 (fair-esm)	2.0	Protein language model for sequence embeddings
scikit-learn	1.4+	XGBoost/MLP classifiers, calibration, evaluation metrics
SQLAlchemy	2.0+	ORM for PostgreSQL interaction
Celery	5.3+	Async task queue for long-running jobs
Apache Airflow	2.8+	Data pipeline orchestration and scheduling
BioPython	1.83+	Protein sequence parsing, PDB file handling
pandas / numpy	Latest stable	Data manipulation and numerical computation

4.2.2 Frontend Technologies

Technology	Version	Purpose
React	18+	SPA framework — component-based UI
TypeScript	5.0+	Type-safe frontend development
Mol*	Latest	Primary WebGL 3D protein structure viewer
3Dmol.js	2.0+	Alternative/fallback WebGL viewer for simpler rendering cases
D3.js	7.0+	Network graph rendering for Interaction Network Viewer
Recharts	2.0+	Expression heatmap and calibration curve visualizations
TailwindCSS	3.0+	Utility-first styling
Zustand	4.0+	Lightweight state management
React Query	5.0+	Data fetching, caching, and WebSocket integration

4.2.3 Database Management System

System	Purpose	Justification
PostgreSQL 15	Primary relational database	ACID compliance, excellent JSON support, proven at scale
Redis 7	Embedding cache + task queue broker	Sub-millisecond reads for cached embeddings
S3-compatible Object Storage	PDB files, model checkpoints, report PDFs	Cost-effective for large binary files; CDN-ready

4.2.4 Configuration and Environment Management

Tool	Purpose
Docker + Docker Compose	Containerization and local development environment
Kubernetes (K8s)	Production orchestration and horizontal scaling
GitHub Actions	CI/CD pipeline — automated testing, linting, Docker builds
Prometheus + Grafana	System monitoring and alerting
Sentry	Error tracking and performance monitoring

4.2.5 Architectural Trade-offs & Strategic Decisions

Decision	Chosen Approach	Rationale
Frontend framework	React + TypeScript over Vue/Svelte	Larger ecosystem; better Mol* integration examples available
Backend framework	FastAPI over Django	Async-native; auto-OpenAPI docs; better for ML service integration
3D Viewer	Mol* primary, 3Dmol.js fallback	Mol* is used on PDB.org and EBI; 3Dmol.js as proven lighter fallback
ML Framework	PyTorch over TensorFlow	ESM-2 is distributed as a PyTorch model; ecosystem alignment
Database	PostgreSQL over NoSQL	Structured relationships (proteins → annotations → interactions) benefit from relational model
Task Queue	Celery + Redis over cloud-native queues	Avoids vendor lock-in; self-hosted Redis doubles as embedding cache

4.3 Use Case Specifications

4.3.1 Researcher — Protein Function Query

Attribute	Description
Use Case ID	UC-001
Name	Query Protein Function by UniProt ID or Sequence
Actor	Biomedical Researcher
Preconditions	User is authenticated; Model 1 is deployed; protein exists in database or can be fetched from UniProt
Trigger	User enters a UniProt ID (e.g., P01308) or raw amino acid sequence in the dashboard search bar
Main Flow	1. User enters UniProt ID or sequence2. System fetches protein metadata from database/UniProt API3. System retrieves or computes ESM-2 embeddings4. Model 1 generates GO MF predictions with confidence scores5. System generates residue-level attention heatmap6. Dashboard renders: function prediction report + Sequence Annotator + 3D Viewer with highlighted residues7. Evidence Graph is assembled and displayed8. User can expand any GO term to see supporting database references
Alternate Flow	If protein not in AlphaFold DB: SWISS-MODEL fallback is triggered; user sees a 'Building homology model...' progress indicator
Postconditions	User has a complete, evidence-backed function prediction with visualizations and references
Exceptions	Invalid UniProt ID: system returns descriptive error suggesting the user verify the ID on UniProt directly

4.3.2 Researcher — 3D Structure Exploration (Standalone)

Attribute	Description
Use Case ID	UC-002
Name	Explore 3D Protein Structure Interactively
Actor	Biomedical Researcher / Graduate Student
Preconditions	User has a UniProt ID, PDB ID, or PDB file; WebGL is supported by the browser
Trigger	User opens '3D Structure Viewer' tool from the dashboard
Main Flow	1. User enters PDB ID, UniProt ID (mapped to AlphaFold accession), or uploads a PDB file2. System fetches structure from PDB, AlphaFold DB, or

	accepts upload3. Structure is rendered in the browser using Mol* with default cartoon representation4. User can switch representation (cartoon/surface/ball-and-stick), color by pLDDT or B-factor, highlight residues5. User can export the current view as a PNG image
Postconditions	User has an interactive 3D protein structure view without any desktop software installation
Exceptions	Structure file too large (>50MB): user is warned and offered a simplified rendering mode

4.3.3 Researcher — Protein-Protein Interaction Prediction

Attribute	Description
Use Case ID	UC-005
Name	Predict Protein-Protein Interaction
Actor	Biomedical Researcher
Preconditions	Model 2 is deployed; both proteins exist in database or can be fetched
Trigger	User enters two UniProt IDs in the PPI prediction tool
Main Flow	1. User enters UniProt ID A and UniProt ID B2. System fetches or retrieves ESM-2 embeddings for both proteins3. Paired embeddings (concatenation, difference, product) are computed4. Model 2 outputs: interaction probability, interaction type (activating/inhibitory), predicted interaction residues5. Co-visualization renders both proteins with binding interface highlighted6. Interaction Network Viewer shows the pair in broader PPI network context7. DrugBank is queried for compounds modulating this interaction8. Full structured report is assembled and displayed
Postconditions	User has a complete PPI prediction with structural visualization, network context, and pharmacological relevance
Exceptions	If one or both proteins are not in training distribution, a distribution shift warning is shown

4.3.4 Clinical Scientist — CCA Biomarker Candidate Report

Attribute	Description
Use Case ID	UC-009
Name	Generate CCA Biomarker Candidate Report
Actor	Clinical Scientist

Preconditions	All three models are deployed; TCGA CCA expression data is loaded; user has Clinical Scientist role
Trigger	User opens the 'CCA Biomarker Dashboard' from the dashboard
Main Flow	1. System loads pre-computed differential expression results for CCA (tumor vs. normal)2. User can filter by fold-change threshold and statistical significance (p-value cutoff)3. System runs candidate proteins through the full pipeline: function prediction → PPI mapping → essentiality analysis → mutation characterization → pharmacological cross-referencing4. Results are ranked by a composite evidence score5. User can click any candidate to view its full prediction report6. User can export the ranked list and individual reports as a structured PDF
Postconditions	User has a ranked, evidence-backed list of CCA biomarker and drug target candidates ready for experimental validation planning
Exceptions	If TCGA data is not loaded (e.g., data pipeline not run): system displays a clear status message with instructions

4.3.5 Platform Administrator — Data Pipeline Management

Attribute	Description
Use Case ID	UC-011
Name	Re-run Data Pipeline on New UniProt Release
Actor	Platform Administrator
Preconditions	New UniProt release is available; admin has pipeline execution permissions
Trigger	Admin triggers pipeline run from admin dashboard or schedules it via Airflow DAG
Main Flow	1. Admin initiates pipeline run (manual trigger or scheduled)2. Pipeline fetches latest Swiss-Prot reviewed entries3. New entries are diffed against existing database; new proteins are added, updated entries are versioned4. New GO annotations are fetched and filtered to experimental evidence codes5. Dataset splits are recomputed (time-based)6. New data card is generated with version, date, and change summary7. Admin receives completion notification with summary statistics
Postconditions	Database reflects the latest UniProt release; all artifacts are versioned; models can be retrained on new data
Exceptions	API failure during ingestion: pipeline marks failed step, sends alert, and retries with exponential backoff. Partial failures do not corrupt existing data.

4.4 Use Case Diagrams

The following sections describe the major actors and their interactions with ProtMind system modules. Full diagrammatic representations are to be produced using draw.io or Lucidchart during Phase 2 implementation.

4.4.1 Researcher Module

Actor	Use Cases
Researcher	UC-001: Query protein function, UC-002: Explore 3D structure, UC-003: Compare structures, UC-004: View annotated sequence, UC-005: Predict PPI, UC-006: Explore interaction network, UC-007: Analyze mutation effect, UC-008: View expression heatmap, UC-010: Export PDF report

4.4.2 Clinical Scientist Module

Actor	Use Cases
Clinical Scientist	UC-009: Generate CCA biomarker candidate report, UC-008: View expression heatmap, UC-010: Export PDF report, UC-002: Explore 3D structure (for candidate review)

4.4.3 Platform Administrator Module

Actor	Use Cases
Platform Admin	UC-011: Re-run data pipeline, UC-012: Monitor system health and pipeline status, UC-013: Manage user access and roles, UC-014: View audit logs

4.5 Entity Relationship Diagram (ERD)

4.5.1 Introduction

The ProtMind data model is centered on the Proteins entity, which anchors all downstream data: annotations, interactions, structural files, dataset splits, and prediction results. The schema is designed to support both the data pipeline (write-heavy, batch operations) and the dashboard (read-heavy, real-time queries).

4.5.2 ERD Design Approach

- Normalization: The schema is normalized to 3NF to avoid redundancy and ensure consistency across pipeline updates.
- Versioning: Annotation and interaction tables include source and date fields to support time-based splitting and provenance tracking.
- Extensibility: The schema is designed to accommodate new data sources (e.g., additional interaction databases) without structural changes.

4.5.3 List of All Entities

Entity	Description
Proteins	Core entity — one row per UniProt reviewed entry
Annotations	GO term annotations with evidence codes and dates — linked to Proteins
Interactions	PPI data from BioGRID and STRING — linked to two Proteins entries
Structures	Structure file references (PDB, AlphaFold) — linked to Proteins
Embeddings	Cached ESM-2 embeddings — linked to Proteins
Dataset_Splits	Train/validation/test partition assignments — linked to Proteins
Predictions	Model outputs (function, PPI, essentiality) — linked to Proteins
Evidence_Items	Evidence Graph nodes — linked to Predictions and source databases
Expression_Data	TCGA expression values — linked to Proteins and tissue/condition metadata
Pipeline_Runs	Data pipeline execution log with version and statistics
Users	Platform users with roles (Researcher, Clinical Scientist, Admin)
Sessions	User session management with refresh token references

4.5.4 Tables of Entities and Attributes

4.5.4.1 Proteins

Column	Type	Description	Constraints
uniprot_id	VARCHAR(20)	UniProt accession number	PK, NOT NULL, UNIQUE
entry_name	VARCHAR(100)	UniProt entry name (e.g., INS_HUMAN)	NOT NULL
sequence	TEXT	Full amino acid sequence	NOT NULL
organism	VARCHAR(200)	Scientific organism name	NOT NULL
taxon_id	INTEGER	NCBI taxonomy ID	NOT NULL
reviewed	BOOLEAN	TRUE = Swiss-Prot reviewed	NOT NULL, DEFAULT TRUE
protein_name	TEXT	Full protein name from UniProt	

gene_names	TEXT	Gene symbol(s), semicolon-separated	
sequence_length	INTEGER	Number of amino acid residues	NOT NULL
created_at	TIMESTAMP	Date first added to database	NOT NULL
updated_at	TIMESTAMP	Date last updated	NOT NULL

4.5.4.2 Annotations

Column	Type	Description	Constraints
annotation_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL
go_id	VARCHAR(15)	GO term accession (e.g., GO:0005515)	NOT NULL
go_aspect	CHAR(1)	F=Molecular Function, P=Biological Process, C=Component	NOT NULL
evidence_code	VARCHAR(10)	Evidence code (EXP, IDA, IPI, IMP, IGI, IEP)	NOT NULL
annotation_date	DATE	Date annotation was added to GOA	NOT NULL
source_db	VARCHAR(50)	Source database (GOA, UniProt)	NOT NULL

4.5.4.3 Interactions

Column	Type	Description	Constraints
interaction_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id_a	VARCHAR(20)	First protein in interaction	FK → Proteins, NOT NULL
uniprot_id_b	VARCHAR(20)	Second protein in interaction	FK → Proteins, NOT NULL
interaction_type	VARCHAR(50)	Type: physical, genetic, co-expression, etc.	NOT NULL
effect	VARCHAR(50)	Effect: activating, inhibitory, unknown	
confidence_score	FLOAT	STRING/BioGRID confidence score (0–1)	
source_db	VARCHAR(50)	BioGRID or STRING	NOT NULL
pmid	VARCHAR(20)	PubMed reference if available	

4.5.4.4 Structures

Column	Type	Description	Constraints
structure_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL
source	VARCHAR(20)	PDB, AlphaFold, SWISS-MODEL	NOT NULL
accession	VARCHAR(50)	PDB ID or AlphaFold accession	NOT NULL
file_path	TEXT	Object storage path to PDB/mmCIF file	NOT NULL
plddt_mean	FLOAT	Mean pLDDT score (AlphaFold only)	
resolution	FLOAT	Resolution in Angstroms (PDB experimental only)	
fetched_at	TIMESTAMP	When structure was last fetched	NOT NULL

4.5.4.5 Embeddings

Column	Type	Description	Constraints
embedding_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL, UNIQUE per model version
model_version	VARCHAR(50)	ESM-2 model version used	NOT NULL
embedding_path	TEXT	Object storage path to serialized embedding tensor	NOT NULL
computed_at	TIMESTAMP	Computation timestamp	NOT NULL

4.5.4.6 Dataset_Splits

Column	Type	Description	Constraints
split_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL
split_version	VARCHAR(50)	Dataset split version identifier	NOT NULL

partition	VARCHAR(20)	train, validation, or test	NOT NULL
cutoff_date	DATE	Annotation date cutoff for this partition	NOT NULL

4.5.4.7 Predictions

Column	Type	Description	Constraints
prediction_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL
model_type	VARCHAR(20)	function, ppi, essentiality, mutation	NOT NULL
model_version	VARCHAR(50)	Model version that generated this prediction	NOT NULL
input_hash	VARCHAR(64)	SHA-256 of input to enable result caching	NOT NULL
output_json	JSONB	Full structured prediction output	NOT NULL
created_at	TIMESTAMP	Prediction generation timestamp	NOT NULL

4.5.4.8 Expression_Data

Column	Type	Description	Constraints
expression_id	BIGSERIAL	Surrogate primary key	PK
uniprot_id	VARCHAR(20)	Reference to Proteins	FK → Proteins, NOT NULL
tcga_project	VARCHAR(50)	TCGA project code (e.g., TCGA-CHOL for CCA)	NOT NULL
tissue_type	VARCHAR(100)	Tissue or condition label	NOT NULL
sample_type	VARCHAR(50)	tumor, normal, metastatic	NOT NULL
expression_value	FLOAT	Normalized expression value (TPM or FPKM)	NOT NULL
statistical_method	VARCHAR(50)	Normalization method used	NOT NULL

4.5.5 Relationships Between Entities

Entity A	Relationship	Entity B	Cardinality

Proteins	has many	Annotations	1 : N
Proteins	has many	Structures	1 : N
Proteins	has one	Embeddings (per model version)	1 : N
Proteins	has many	Dataset_Splits	1 : N
Proteins	has many	Predictions	1 : N
Proteins	has many	Expression_Data	1 : N
Proteins (A & B)	participates in	Interactions	N : M (via Interactions table)
Predictions	has many	Evidence_Items	1 : N

4.6 Activity Diagrams & Workflows

The following workflows describe the key operational flows within ProtMind. Each activity diagram is documented in structured form, suitable for conversion to UML activity diagrams during implementation.

4.6.1 Activity 1: Protein Sequence Input & Structure Retrieval

Step	Actor	Action	Decision / Branch
1	User	Enter UniProt ID or raw amino acid sequence in dashboard search	—
2	System	Validate input: is it a valid UniProt ID or sequence format?	Invalid → return error message
3	System	Query local database for cached protein entry	Not found → fetch from UniProt API
4	System	Check if AlphaFold DB structure exists for this protein	Not found → trigger SWISS-MODEL fallback
5a	System (AlphaFold path)	Fetch predicted structure + pLDDT scores + PAE matrix	—
5b	System (SWISS-MODEL path)	Submit homology modeling request; poll for completion	Timeout → return partial result with warning
6	System	Store structure in object storage; update Structures table	—

7	Frontend	Render 3D structure in Mol* viewer with pLDDT coloring	—
8	Frontend	Display PAE plot alongside structure	—

4.6.2 Activity 2: Protein Function Prediction

Step	Actor	Action	Decision / Branch
1	User	Submit function prediction request (UniProt ID or sequence)	—
2	System	Check Embeddings cache for existing ESM-2 embedding	Not found → compute embedding via ESM-2 model
3	System	Run multi-label GO MF classifier on embedding	—
4	System	Generate confidence scores per GO term	Score < 0.3 → flag as low-confidence
5	System	Extract residue-level attention weights from ESM-2	—
6	System	Fetch supporting GO term metadata and database references	—
7	System	Assemble Evidence Graph (sequence + structural + literature evidence)	—
8	System	Format output: summary + steps + technical details + references	—
9	Frontend	Render function prediction report, Sequence Annotator, 3D Viewer	—

4.6.3 Activity 3: PPI Prediction

Step	Actor	Action	Decision / Branch
1	User	Enter two UniProt IDs in PPI prediction tool	—
2	System	Retrieve or compute ESM-2 embeddings for both proteins	—
3	System	Compute paired representation (concatenation + difference + product)	—

4	System	Run PPI classifier; output: interaction probability + type	Probability < 0.5 → classify as non-interacting
5	System	Identify predicted interaction residues	—
6	System	Fetch both structures; render co-visualization with interface highlighted	—
7	System	Load Interaction Network Viewer with both proteins highlighted	—
8	System	Query DrugBank for compounds modulating this interaction	No results → display 'No known modulators found'
9	System	Assemble full PPI report	—

4.6.4 Activity 4: Mutation Effect Analysis

Step	Actor	Action	Decision / Branch
1	User	Select a protein and specify a point mutation (e.g., A23V)	—
2	System	Retrieve wild-type sequence and structure	—
3	System	Generate mutant sequence by applying substitution	—
4	System	Compute ESM-2 embedding for mutant sequence	—
5	System	Run mutation effect classifier: neutral / mildly / severely disruptive	—
6	System	Check if mutation position is in active site, allosteric site, or drug-binding region	—
7	System	Request mutant structure via AlphaFold or SWISS-MODEL	—
8	System	Compute RMSD between wild-type and mutant structures	—
9	Frontend	Render Structure Comparator overlay; highlight mutation position in Sequence Annotator	—
10	System	Assemble mutation analysis report with clinical significance flags	—

4.6.5 Activity 5: CCA Biomarker Pipeline

Step	Actor	Action	Decision / Branch
1	System	Load TCGA CCA expression data (tumor vs. normal)	Data not loaded → admin alert
2	User	Set filter thresholds (fold-change cutoff, p-value cutoff)	—
3	System	Identify significantly differentially expressed proteins meeting criteria	No candidates → relax thresholds and notify user
4	System	Run all candidates through full pipeline: function → PPI → essentiality → mutation → pharmacology	—
5	System	Score each candidate using composite evidence score	—
6	System	Rank candidates; flag top candidates with high pharmacological relevance	—
7	Frontend	Render CCA Biomarker Dashboard with ranked list	—
8	User	Click any candidate to expand full prediction report	—
9	User	Optionally export ranked list + reports as structured PDF	—

4.6.6 Activity 6: Data Pipeline Execution

Step	Actor	Action	Decision / Branch
1	Admin / Scheduler	Trigger pipeline run (manual or scheduled via Airflow DAG)	—
2	Pipeline	Fetch latest Swiss-Prot reviewed entries from UniProt API	API failure → retry with backoff; alert if > 3 failures
3	Pipeline	Diff new entries against existing Proteins table	—
4	Pipeline	Insert new proteins; version-update changed entries	—
5	Pipeline	Fetch GO annotations from GOA; filter to experimental evidence codes	—
6	Pipeline	Update Annotations table; recompute dataset splits (time-based)	—

7	Pipeline	Fetch updated BioGRID and STRING interactions	—
8	Pipeline	Generate new data card with version, date, change summary	—
9	System	Send completion notification to admin with statistics	—

4.7 Data Flow Diagrams (DFD)

4.7.1 Introduction

The ProtMind DFDs describe how data flows between external entities, system processes, and data stores at three levels of abstraction: context (Level 0), main processes (Level 1), and detailed prediction pipeline (Level 2).

4.7.2 DFD Level 0 — Context Diagram

External Entity	Data Flowing INTO System	Data Flowing OUT OF System
Researcher/User	UniProt IDs, sequences, PDB IDs, query text, mutation specifications	Prediction reports, 3D visualizations, evidence graphs, exported PDFs
Platform Admin	Pipeline trigger commands, configuration updates	System health status, pipeline completion reports, audit logs
UniProt API	Protein sequences, annotations, feature data	Data ingestion requests
AlphaFold DB	Predicted structures, pLDDT scores, PAE matrices	Structure fetch requests
SWISS-MODEL	Homology model structures	Modeling requests (sequence + template)
PDB	Experimental structures	Structure fetch requests
STRING/BioGRID	Interaction network data	Interaction data requests
DrugBank	Drug-target association data	Target query requests
TCGA / NCI GDC	CCA expression profiles	Data ingestion requests (batch, Phase 1)

4.7.3 DFD Level 1 — Main Processes

Process	Inputs	Outputs	Data Stores Read/Written

P1: Data Ingestion Pipeline	API data from external sources	Normalized records	Proteins, Annotations, Interactions, Structures, Expression_Data
P2: User Query Processing	User input (natural language or structured)	Structured API request	Sessions
P3: Structure Retrieval & Rendering	UniProt ID / PDB ID / uploaded file	Structure file + pLDDT data	Structures (read/write)
P4: ML Inference	Protein sequence / embedding	Prediction output + confidence scores	Embeddings (read/write), Predictions (write)
P5: Explainability & Evidence Assembly	Prediction output + source data	Evidence Graph + formatted report	Annotations, Structures (read)
P6: Visualization Rendering	Structure data, annotation data, expression data	Rendered UI components (3D, sequence, heatmap, network)	All data stores (read)
P7: Report Export	Assembled report content	PDF report file	Predictions (read)

4.7.4 DFD Level 2 — Detailed ML Inference Pipeline

Sub-Process	Input	Output	Notes
P4.1: Embedding Retrieval	UniProt ID	ESM-2 embedding tensor	Cache check first; compute if miss
P4.2: Function Classification	ESM-2 embedding	GO MF terms + confidence scores	Multi-label classifier output
P4.3: Attention Extraction	ESM-2 model + sequence	Residue-level attention weights	Used for Sequence Annotator heatmap
P4.4: PPI Pair Encoding	Two ESM-2 embeddings	Paired representation vector	Concat + diff + product
P4.5: PPI Classification	Paired representation	Interaction probability + type + residues	Binary + multi-class output
P4.6: Mutation Encoding	WT and mutant ESM-2 embeddings	Delta embedding representation	Captures sequence change signal

P4.7: Essentiality/Mutation Scoring	Delta embedding	Impact score + essentiality score	Cross-species PLM output

4.8 Prototype Overview

The following sections describe the key interface views of ProtMind as designed during the prototype phase. Full interactive prototypes are to be developed in Figma during Phase 2.

4.8.1 Landing Dashboard

UI Element	Description
Header	ProtMind logo, navigation links, user account menu
Central Search Bar	Prominent search input: 'Enter a UniProt ID, sequence, or ask a question...'
Quick Access Tiles	Six tiles — one per visualization tool — with icons and one-line descriptions
Recent Queries	Last 5 queries made by the authenticated user, clickable to restore results
Status Bar	Current database version, last pipeline run date, model versions deployed

4.8.2 3D Structure Viewer View

UI Element	Description
Input Panel (left)	Text input for UniProt ID or PDB ID; file upload button; 'Load Structure' button
3D Canvas (center)	Full Mol* viewer with rotation, zoom, pan controls; representation selector
Toolbar (top of canvas)	Representation mode (cartoon/surface/sticks), color scheme selector, export PNG button
pLDDT Legend (right)	Color scale: blue (very high >90) → yellow (low 50–70) → orange/red (<50)
PAE Plot (bottom right)	Interactive PAE heatmap; click region to highlight corresponding residues in 3D viewer
Residue Info Panel	Click any residue in 3D to show: position, amino acid, pLDDT score, nearby annotations

4.8.3 Function Prediction Report View

UI Element	Description

Summary Box (top)	Plain-language summary of top predicted GO term(s) in non-technical language
GO Term Table	Ranked list of predicted GO MF terms with confidence bars and evidence code links
Sequence Annotator Panel	Annotated sequence with residue-level attention heatmap overlay; GO-relevant regions highlighted
3D Viewer Integration	Linked 3D view with predicted functional residues highlighted in the structure
Evidence Graph Panel	Interactive graph: prediction node → connected to UniProt entries, GO term definitions, literature
Step-by-Step Explanation	Collapsible section: what data was used, what model ran, what the scores mean
References Section	Database and literature citations supporting the prediction

4.8.4 CCA Biomarker Dashboard View

UI Element	Description
Filter Controls (top)	Fold-change slider, p-value cutoff input, tissue type selector
Expression Heatmap (main)	TCGA CCA expression heatmap — rows=proteins, columns=samples; color=expression level
Candidate Ranking Table	Ranked list with: rank, UniProt ID, gene name, fold-change, top GO term, essentiality, drug relevance score
Candidate Detail Panel	Click any row → expand full pipeline report for that protein
Export Controls	Export ranked table as CSV; export full evidence reports as PDF

4.8.5 Structure Comparator View

UI Element	Description
Input Panel	Two structure inputs (UniProt IDs or PDB IDs); 'Compare' button
Split 3D View	Side-by-side or overlaid 3D rendering of both structures; toggle between modes
RMSD Display	Global RMSD score prominently displayed; per-domain RMSD breakdown table
Difference Highlighting	Regions of structural divergence highlighted in red; conserved regions in blue

Mutation Marker	If comparison is WT vs. mutant: mutation position marked with a distinct color

4.9 Tooling & Technical References

4.9.1 System Modeling & Diagramming

Tool	Purpose
draw.io / Lucidchart	Use case diagrams, architecture diagrams, DFDs, activity diagrams
dbdiagram.io	ERD design and relational schema documentation
Figma	UI/UX prototyping and design system
PlantUML	Programmatic UML diagram generation from code

4.9.2 UI/UX Design & Prototyping

Tool	Purpose
Figma	High-fidelity interactive prototypes for all dashboard views
Mol* Viewer	WebGL 3D protein structure rendering (primary)
3Dmol.js	WebGL 3D structure rendering (fallback / lightweight mode)
D3.js	Interaction network graph visualization
Recharts	Expression heatmap, PAE plots, calibration curves

4.9.3 Documentation & Publishing Tools

Tool	Purpose
GitHub / Git	Version control for code, model artifacts, and data cards
Sphinx + ReadTheDocs	API documentation and developer guides
Jupyter Notebooks	Exploratory data analysis, model evaluation, and public demo notebooks
Weights & Biases (W&B)	Model training tracking, hyperparameter logging, and experiment comparison
LaTeX / Overleaf	Technical report and academic publication preparation

4.9.4 Core Technical Stack

Layer	Tool	Version
Frontend Framework	React + TypeScript	18+ / 5.0+
Backend Framework	FastAPI	0.110+
ML / Embeddings	PyTorch + ESM-2 (fair-esm)	2.0+ / 2.0
Database	PostgreSQL	15+
Cache / Queue	Redis + Celery	7.0+ / 5.3+
Pipeline Orchestration	Apache Airflow	2.8+
Containerization	Docker + Kubernetes	24+ / 1.28+
CI/CD	GitHub Actions	—
Monitoring	Prometheus + Grafana	—

4.9.5 External Integration APIs

API	Base URL	Authentication	Rate Limit Strategy
UniProt REST API	https://rest.uniprot.org	None (public)	Respect 1 req/sec guideline; batch where possible
AlphaFold DB API	https://alphafold.ebi.ac.uk/api	None (public)	Cache all structure files locally after first fetch
SWISS-MODEL API	https://swissmodel.expasy.org/api	API key required	Queue requests; poll for completion; cache results
PDB REST API	https://data.rcsb.org/rest/v1	None (public)	Cache structure files locally
STRING API v11.5	https://string-db.org/api	None (public)	Batch requests by protein list
BioGRID API	https://webservice.thebiogrid.org	API key required	Paginated bulk ingestion at pipeline time
DrugBank API	https://go.drugbank.com/releases/latest/downloads	License key required	Query at prediction time; cache results per target

NCI GDC API	https://api.gdc.cancer.gov	None (public, open-access data)	Bulk data download at pipeline time only

Document End

This Software Requirements Specification covers all seven development phases of ProtMind — from data infrastructure through CCA clinical demonstration. All functional requirements, non-functional requirements, system architecture, entity schemas, activity workflows, data flows, and tooling decisions documented herein serve as the authoritative reference for ProtMind development, testing, and validation activities.

Chapter 5: UI/UX Design

This chapter documents the complete UI/UX design specification for ProtMind — covering design philosophy, design system (colors, typography, spacing, components), information architecture, wireframe specifications for all major views, interaction design, user flow diagrams, accessibility guidelines, and responsive design rules. This chapter serves as the authoritative design reference for all frontend development work.

5.1 Design Philosophy

Core Design Principle

ProtMind's interface is built on a single governing principle: scientific clarity over visual complexity. Every design decision — color, spacing, layout, interaction — must serve the researcher's task, not decorate it. The interface should feel like a precision instrument, not a consumer app.

5.1.1 Design Pillars

Pillar	Definition	Design Implication
Clarity	Information is presented without ambiguity. Confidence scores, evidence levels, and data provenance are always visible.	Dense but scannable layouts; no hidden critical information; consistent score visualization
Trust	Every AI output is accompanied by the evidence that supports it.	Evidence Graph always accessible; model version always displayed; calibration indicators on scores

	The system earns trust by showing its work.	
Efficiency	A researcher should reach any insight in the fewest possible interactions.	Search-first design; direct tool access from landing; keyboard shortcuts for power users
Scientific Neutrality	The interface does not celebrate or alarm — it informs. Visual language must not inject bias into scientific interpretation.	Neutral color language except for universal conventions (red = warning, green = validated); no gamification
Cross-Platform Parity	The experience must be identical on macOS, Windows, Linux, and all supported browsers.	CSS-only layouts; no OS-native controls; WebGL rendering tested across all platforms

5.1.2 Design Anti-Patterns (Explicitly Avoided)

- No gamification or engagement mechanics — protein research is not a game.
- No modal overuse — dialogs only for destructive or irreversible actions.
- No dark patterns — no hidden controls, no misleading affordances.
- No decorative animations — transitions serve orientation only, never attention.
- No information hiding behind progressive disclosure when the data is the product.

5.2 Design System

5.2.1 Color Palette

The color palette is anchored in deep research-blue as the primary brand color, with a strictly controlled palette of functional colors. Colors carry meaning — they are never decorative.

Primary Palette

Swatch	Color / Hex	Usage
	Navy Blue (Primary) `#1B4F8A`	Main headings, primary action buttons, brand identity elements, section anchors
	Cobalt Blue (Secondary) `#2E75B6`	Subheadings, active states, hover highlights, borders on key components
	Accent Blue `#0070C0`	Links, interactive labels, third-level headings, clickable element indicators
	Sky Tint (Background) `#EBF3FB`	Callout box backgrounds, table alternating rows, panel backgrounds, card fills

	Mid Blue (Surface) #D0E4F5	Section dividers, phase header backgrounds, selected state fills

Neutral Palette

Swatch	Color / Hex	Usage
	Deep Dark (Body Text) #1A1A2E	All body copy, table cell text, form input text
	Mid Gray (Secondary Text) #666666	Subtitles, metadata labels, placeholder text, footer copy
	Off-White (Alt Row) #F2F7FC	Alternating table rows, subtle background differentiation
	Pure White #FFFFFF	Page background, card backgrounds, input field backgrounds
	Border Gray #CCCCCC	Table borders, divider lines, input field outlines

Semantic / Functional Colors

Swatch	Color / Hex	Usage
	Validated Green #00B050	Experimentally verified annotations, high-confidence predictions (>80%), success states
	Caution Amber #FF9900	Medium-confidence predictions (50–80%), model distribution warnings, caution flags
	Alert Red #C00000	Low-confidence predictions (<50%), severely disruptive mutations, error states
	Pharmacology Purple #7030A0	Drug-related data, DrugBank results, clinical trial indicators
	Deep Green (Conservation) #375623	Structurally conserved regions in comparator, high essentiality scores

pLDDT Confidence Color Scale (AlphaFold Convention)

Swatch	Color / Hex	Usage
	Very High (>90) #0053D6	Highly confident predicted structure regions — reliable for analysis

	Confident (70–90) #65CBF3	Generally accurate predicted regions
	Low (50–70) #FFDB13	Predicted with low confidence — treat with caution
	Very Low (<50) #FF7D45	Likely disordered or unreliable — not suitable for structural interpretation

5.2.2 Typography

Role	Typeface	Size	Weight	Usage
Display Title	Inter	48–72px	700 Bold	Landing page hero text, major feature titles
Page Title (H1)	Inter	32px	700 Bold	Chapter and section titles
Section Header (H2)	Inter	24px	600 SemiBold	Major subsection headers
Sub-Header (H3)	Inter	20px	600 SemiBold	Component group headers, panel titles
Label / H4	Inter	16px	600 SemiBold	Field labels, card headers
Body Text	Inter	15px	400 Regular	Paragraphs, descriptions, explanations
Body Emphasis	Inter	15px	500 Medium	Inline emphasis within body copy
Caption / Meta	Inter	13px	400 Regular	Timestamps, database references, footnotes
Data / Monospace	JetBrains Mono	14px	400 Regular	Sequences, UniProt IDs, PDB accessions, confidence values
Score Display	JetBrains Mono	18px	700 Bold	Confidence scores, RMSD values, Fmax scores — requires precision reading

Inter is chosen as the primary typeface for its exceptional legibility at small sizes, its open-source licensing, and its wide character set including all scientific notation required. JetBrains Mono is used exclusively for technical data strings where character-level precision matters — UniProt IDs must never render ambiguously.

5.2.3 Spacing & Layout Grid

Token	Value	Usage
--space-1	4px	Micro gaps — icon-to-label, tight inline spacing
--space-2	8px	Small gaps — within form fields, compact list items
--space-3	12px	Default component internal padding
--space-4	16px	Standard gap between related elements
--space-6	24px	Section internal spacing, card padding
--space-8	32px	Gap between major sections within a page
--space-12	48px	Gap between page-level sections
--space-16	64px	Hero section vertical padding
--radius-sm	4px	Button corners, small badge corners
--radius-md	8px	Card corners, panel corners, input fields
--radius-lg	12px	Modal corners, large card corners
--shadow-sm	0 1px 3px rgba(0,0,0,0.08)	Resting card elevation
--shadow-md	0 4px 12px rgba(0,0,0,0.12)	Hover card elevation, active panel
--shadow-lg	0 8px 32px rgba(0,0,0,0.16)	Modal elevation, floating toolbar

Layout Grid

Breakpoint	Grid Columns	Gutter	Margin	Notes
Mobile (< 768px)	4 columns	16px	16px	Visualization tools show simplified 2D fallback
Tablet (768–1199px)	8 columns	24px	24px	Sidebar panels collapse to bottom drawers
Desktop (1200–1599px)	12 columns	24px	32px	Standard layout — main target viewport
Wide (1600px+)	12 columns (max 1440px wide)	32px	Auto	Dashboard max-width capped at 1440px, centered

5.2.4 Iconography & Visual Language

Icon Set	Usage	Source
Phosphor Icons	Primary icon set — navigation, actions, status indicators	phosphoricons.com — MIT licensed
Custom Bioinformatics Icons	Protein helix, DNA strand, molecule node, AlphaFold badge	Designed in Figma; exported as SVG
Evidence Level Badges	EXP / IDA / IMP / etc. — colored pill badges for GO evidence codes	Custom — color-coded by reliability tier
Confidence Meter	Horizontal progress bar with color-coded fill (green/amber/red)	Custom React component
pLDDT Color Scale	Color-coded residue confidence in all structure views	Matches AlphaFold DB convention exactly

5.3 Component Library

ProtMind's component library is built in React with TypeScript and styled using TailwindCSS utility classes. Every component is documented with its props, variants, states, and accessibility requirements.

5.3.1 Core Interactive Components

◆ **SearchBar** — The central interaction element — accepts UniProt IDs, sequences, or natural language

Property	Value / Spec	Notes
`variant`	'hero' | 'compact'	*hero: landing page large format; compact: embedded in tool panels*
`placeholder`	string	*Adapts per context: 'Enter UniProt ID, sequence, or ask...'*
`onSearch(query)`	function	*Called on Enter or search button click*
`isLoading`	boolean	*Shows animated spinner inside input during fetch*
`suggestions`	string[]	*Inline autocomplete from UniProt ID prefix lookup*
`a11y`	role='searchbox', aria-label, keyboard nav	*Full keyboard support; escape clears; up/down for suggestions*

◆ **ConfidenceScore** — Displays a model confidence value with color coding and context label

Property	Value / Spec	Notes

`value`	number (0–1)	*Raw confidence probability — rendered as percentage*
`label`	string	*e.g. 'GO:0005515 — Protein binding'*
`size`	'sm' | 'md' | 'lg'	*Controls text and bar sizing*
`showBar`	boolean	*Whether to show horizontal fill bar beneath percentage*
`colorRule`	'plddt' | 'prediction' | 'interaction'	*Determines color thresholds — pLDDT uses AlphaFold scale; prediction uses 80/50 thresholds*

◆ EvidenceBadge — GO annotation evidence code pill badge

Property	Value / Spec	Notes
`code`	'EXP' | 'IDA' | 'IPI' | 'IMP' | 'IGI' | 'IEP'	*Evidence code from GOA*
`tooltip`	string	*Full evidence code description shown on hover*
`variant`	'filled' | 'outlined'	*filled: strong emphasis; outlined: secondary context*
`color`	Auto from code	*EXP=deep green; IDA=green; IPI=teal; IMP=blue; IGI=mid-blue; IEP=gray-blue*

◆ StructureViewer — The core 3D rendering component wrapping Mol*

Property	Value / Spec	Notes
`structureUrl`	string (URL to PDB/mmCIF)	*Loaded on mount or when URL changes*
`colorScheme`	'plddt' | 'bfactor' | 'secondary' | 'residue'	*Controls coloring mode in Mol**
`representation`	'cartoon' | 'surface' | 'ball-stick'	*Defaults to cartoon*
`highlightedResidues`	number[]	*Array of residue indices to highlight (from prediction)*
`onResidueClick(idx)`	function	*Called when user clicks a residue in the viewer*
`showPAE`	boolean	*Whether to render the PAE heatmap panel alongside the structure*
`exportEnabled`	boolean	*Shows export PNG button in viewer toolbar*

◆ InteractionGraph — D3.js-based PPI network graph

Property	Value / Spec	Notes
`nodes`	ProteinNode[]	*Each node: { id, uniprotId, name, isHighlighted }*

`edges`	InteractionEdge[]	*Each edge: { source, target, type, confidence, source_db }*
`centerNodeId`	string	*UniProt ID of the protein to center the layout on*
`onNodeClick(id)`	function	*Expands node neighborhood or opens protein detail panel*
`onEdgeClick(edge)`	function	*Opens edge detail panel with interaction data + drug context*
`filterMinConfidence`	number (0–1)	*Hides edges below this STRING/BioGRID confidence threshold*

◆ SequenceAnnotator — Scrollable annotated amino acid sequence display

Property	Value / Spec	Notes
`sequence`	string	*Full amino acid sequence*
`annotations`	Annotation[]	*Array of { start, end, type, label, color } regions*
`attentionWeights`	number[]	*Per-residue attention values — rendered as heatmap overlay*
`highlightedPositions`	number[]	*Specific positions to mark (e.g., mutation sites)*
`onResidueHover(idx)`	function	*Shows tooltip with residue metadata on hover*
`colorMode`	'annotation' | 'attention' | 'both'	*Controls what is shown as color overlay*

5.3.2 Layout Components

Component	Description	Key Props
DashboardShell	Root layout: global header, left nav, main content area, right context panel	activeView, onNavigate, user, notifications
SplitPanel	Resizable two-panel layout (e.g., 3D viewer + sequence annotator side-by-side)	leftContent, rightContent, defaultSplit, minLeftWidth
ReportCard	Structured output container for AI prediction results	title, summary, steps, details, references, exportEnabled
EvidenceGraph	Interactive network of prediction → supporting evidence nodes	predictionId, depth (1–3 hops), onNodeClick
ToolTile	Dashboard landing tile for direct tool access	toolId, icon, title, description, onClick, badge (e.g., 'New')

ProgressJob	WebSocket-connected job status display for long-running tasks	jobId, onComplete, onError, stages (string[])
FilterBar	Horizontal controls for threshold sliders, dropdown filters, search inputs	filters (FilterConfig[]), onChange, onReset
DataTable	Virtualized scrollable data table for protein lists and candidate rankings	columns, rows, sortable, onRowClick, exportCSV

5.3.3 States & Variants for All Components

State	Visual Treatment	Examples
Default / Resting	Standard colors, 1px border, --shadow-sm elevation	All components at rest
Hover	Border color → C.mid (Cobalt Blue), --shadow-md elevation, cursor: pointer	Cards, tiles, table rows, graph nodes
Active / Selected	Fill → C.lightBg, border → C.blue, bold label	Selected table row, active nav item, expanded panel
Loading	Pulsing skeleton animation (#E8EEF6 → #D0DCF0 cycle), 1.2s duration	Structure viewer while loading, prediction score while computing
Error	Red left border (--color-alert), error icon, descriptive message text	Invalid UniProt ID, API timeout, model failure
Empty	Centered neutral icon + guidance text (no red — not an error)	No candidates found, empty interaction network
Disabled	40% opacity, cursor: not-allowed, no hover effect	Buttons for features not yet in current phase
High Confidence	Green accent, validated badge	GO predictions >80%, EXP evidence code
Low Confidence	Amber accent, caution icon	Predictions 50–80%, SWISS-MODEL fallback result
Very Low Confidence	Red accent, warning icon	Predictions <50%, severely disruptive mutation

5.4 Information Architecture

5.4.1 Navigation Structure

Level	Section	Access	Notes

Global Nav (top)	Dashboard, Search, Tools, Reports, Admin	Always visible	Collapses to icon-only on tablet
Left Sidebar	Tool-specific sub-navigation (context-sensitive)	Visible in tool views	Hidden on landing dashboard
Tool Tiles (landing)	6 direct tool tiles + AI query bar	1 click from landing	Most common entry point
Breadcrumb	Dashboard > Tool Name > Protein ID	All non-landing views	Enables multi-level back navigation
Right Context Panel	Protein metadata, evidence summary, related links	Auto-populated from active query	Collapsible; 300px default width

5.4.2 Site Map

Route	View	Accessible To
/	Landing Dashboard (search + tool tiles + recent queries)	All authenticated users
/search	Protein Search Results	All authenticated users
/protein/:uniprotId	Protein Overview Page	All authenticated users
/tools/structure-builder	Sequence-to-Structure Builder	All authenticated users
/tools/structure-viewer	3D Structure Viewer (Standalone)	All authenticated users
/tools/sequence-annotator	Sequence Annotator (Standalone)	All authenticated users
/tools/structure-comparator	Structure Comparator (Standalone)	All authenticated users
/tools/interaction-network	Interaction Network Viewer (Standalone)	All authenticated users
/tools/expression-heatmap	Expression Heatmap (Standalone)	All authenticated users
/predict/function	Protein Function Prediction (AI)	All authenticated users
/predict/ppi	PPI Prediction (AI)	All authenticated users
/predict/mutation	Mutation Effect Analysis (AI)	All authenticated users
/predict/essentiality	Cross-Species Essentiality (AI)	All authenticated users
/cca	CCA Biomarker Dashboard	Researcher + Clinical Scientist roles
/reports/:reportId	Saved Prediction Report View	Report owner

/admin	Admin Dashboard (pipeline, users, health)	Platform Admin role only
/admin/pipeline	Data Pipeline Management	Platform Admin role only

5.5 Wireframe Specifications

The following wireframe specifications document every major view in ProtMind. Each specification describes the zone layout, component placement, and interaction behavior. Full high-fidelity wireframes are to be produced in Figma using these specs as the source of truth.

5.5.1 Landing Dashboard

⬜ **Landing Dashboard — /**

Entry point for all users. Optimized for the two primary interaction modes: search and direct tool access.

- **Top Bar (64px):** Logo (left) · Global navigation links (center) · User avatar + notification bell (right). Background: white, bottom border: 1px C.mid.
- **Hero Zone (240px):** Centered SearchBar (hero variant, 640px max-width). Below: 3 example queries as clickable chips ('What does P01308 do?', 'Compare P01308 and P01315', 'Find CCA drug targets'). Background: linear gradient from white → C.lightBg.
- **Tool Tiles Grid (auto):** 2-column × 3-row grid of ToolTile components (6 tiles). Each tile: 200px tall, icon (48px, C.accent), tool name (H4), one-line description, 'Open Tool' ghost button. Tiles have --shadow-sm; hover → --shadow-md + blue top border.
- **Recent Queries (auto):** Section header 'Recent Queries'. List of last 5 queries: protein name · query type badge · timestamp · confidence score if available. Click to restore result. Empty state: 'No recent queries — try searching above'.
- **System Status Bar (40px):** Bottom of page: Database version · Last pipeline run · Models deployed (version badges). Background: C.lightBg, 12px font, gray text.

5.5.2 3D Structure Viewer

⬜ **3D Structure Viewer — /tools/structure-viewer**

Web-based full-feature equivalent of UCSF Chimera. Standalone tool — no AI required.

- **Top Bar:** Shared global nav + breadcrumb: Dashboard > 3D Structure Viewer.
- **Input Panel (left, 280px):** Input mode tabs: 'PDB ID' | 'UniProt ID' | 'Upload File'. Text input or file dropzone. Representation selector (Cartoon / Surface / Ball-Stick). Color scheme selector (pLDDT / B-factor / Secondary Structure / Chain). 'Load Structure' button (primary). Residue Info accordion: expands on residue click.

- **3D Canvas (center, fill):** Full Mol* viewer. Toolbar overlay (top): zoom controls, reset view, export PNG. Bottom-left overlay: protein name + UniProt ID + resolution/pLDDT mean. Loading state: centered spinner with 'Fetching structure...' message.
- **pLDDT Legend (canvas overlay, bottom-right):** Vertical color scale: Very High (>90) · Confident (70–90) · Low (50–70) · Very Low (<50). Only shown when color scheme = pLDDT.
- **PAE Plot Panel (right, 320px, collapsible):** PAE matrix heatmap (NxN where N = sequence length). Interaction: click a cell → highlights corresponding residues in 3D viewer. Header: 'Predicted Aligned Error'. Info tooltip explaining PAE interpretation.

5.5.3 Protein Function Prediction Report

⬜ Function Prediction Report — /predict/function

AI-generated prediction output. Combines model results with integrated visualizations.

- **Top Bar + Breadcrumb:** Dashboard > Function Prediction > [UniProt ID]. Right: 'Export PDF' button · 'Share' link button.
- **Summary Card (top, full width):** Protein name + organism (H2). Plain-language summary paragraph (auto-generated). Top 3 predicted GO terms as colored chip badges with confidence meters. Model version + dataset version in small caption.
- **Tabbed Detail Area (full width, below summary):** Tabs: 'Predictions' | 'Sequence View' | '3D Structure' | 'Evidence' | 'How it Works'. Active tab underlined in C.blue. Content area below tabs: renders the relevant sub-component.
- **Tab: Predictions:** DataTable of all predicted GO MF terms: GO ID · Term name · Confidence score (bar + %) · Evidence badge · 'View in GO' link. Sortable by confidence.
- **Tab: Sequence View:** Full SequenceAnnotator component. Toggle: 'Show attention heatmap' / 'Show GO regions' / 'Both'. Residue hover tooltip: amino acid · position · pLDDT · annotations.
- **Tab: 3D Structure:** Embedded StructureViewer (compact mode). Predicted functional residues highlighted in C.accent. Toggle between 'Functional residues' and 'pLDDT coloring' highlight modes.
- **Tab: Evidence:** EvidenceGraph component centered. Legend: Prediction node (blue) → UniProt feature nodes (green) → GO term nodes (teal) → Literature nodes (gray). Depth selector: 1 / 2 / 3 hops.
- **Tab: How it Works:** Step-by-step explanation accordion: Step 1 (embedding) → Step 2 (classification) → Step 3 (attention) → Step 4 (evidence assembly). Each step expandable with technical detail.

5.5.4 PPI Prediction Report

⬜ PPI Prediction Report — /predict/ppi

Two-protein interaction prediction with binding interface visualization and pharmacological context.

- **Top Bar + Input Row:** Two UniProt ID inputs side-by-side with a '↔' icon between them. 'Predict Interaction' button. Shows protein names below inputs once resolved.
- **Result Header Card:** Interaction probability score (large, color-coded). Interaction type badge: 'Activating' (green) / 'Inhibitory' (red) / 'Unknown' (gray). Confidence calibration indicator.
- **Co-Visualization Panel (left 60%):** StructureViewer in co-visualization mode: both proteins rendered simultaneously. Predicted binding interface highlighted in C.accent. Toggle: 'Show protein A only' / 'Show protein B only' / 'Show both'. Residue click → shows interaction evidence tooltip.
- **Interaction Details Panel (right 40%):** Predicted interaction residues list (protein A residues ↔ protein B residues). Confidence per residue pair. Source data: BioGRID / STRING citations.
- **Network Context Section (full width):** InteractionGraph centered on the queried protein pair. Both query proteins highlighted in C.blue. Surrounding nodes: known interaction partners. Filter: by STRING confidence threshold.
- **Pharmacology Section (full width):** DrugBank results table: Drug name · Mechanism · Target (protein A or B or interface) · Phase (preclinical / clinical) · PubMed link. Empty state: 'No known pharmacological modulators found for this interaction.'

5.5.5 Sequence Annotator (Standalone)

⬜ Sequence Annotator — /tools/sequence-annotator

Standalone annotation view — equivalent to UniProt feature viewer but with AI attention overlay.

- **Input Bar (top):** UniProt ID input + 'Load' button. Protein name + organism shown as resolved label once loaded.
- **Annotation Track Selector (left sidebar, 220px):** Checkbox list: GO Functional Regions · Active Sites · Binding Sites · PTM Sites · Signal Peptide · Transmembrane Domains · Mutation Positions · ESM-2 Attention Heatmap. Color swatch per track type.
- **Sequence Display (main area, scrollable horizontal):** Fixed 60-residue-per-line sequence display. Residues colored per active annotation tracks. Line numbers every 10 residues. Hovering a residue shows tooltip: position, amino acid, tracks active at that position.
- **Attention Heatmap Row (below sequence):** When 'ESM-2 Attention Heatmap' is enabled: per-residue bar chart below the sequence. Y-axis: 0–1 normalized attention weight. Color gradient: white → C.accent at max weight.
- **Annotation Legend (bottom):** Color legend for all active tracks. Each entry: color swatch + track name + count of annotated regions.

5.5.6 Structure Comparator

⬜ Structure Comparator — /tools/structure-comparator

WT vs. mutant or cross-species structural comparison with RMSD quantification.

- **Input Row (top):** Structure A: UniProt ID input + label 'Wild-type or Reference'. Structure B: UniProt ID input + label 'Mutant or Comparison'. Optional: mutation specification field (e.g., A23V). 'Compare' primary button.
- **RMSD Summary Bar (full width, below inputs):** Global RMSD value (large monospace display). Interpretation badge: <0.5Å = 'Nearly Identical' (green) / 0.5–2.0Å = 'Minor Differences' (amber) / >2.0Å = 'Significant Differences' (red). Per-domain RMSD breakdown table.
- **3D Comparison View (main, 60% width):** Toggle between: Side-by-Side (two synchronized viewers) / Overlaid (superposed in one viewer). Difference highlighting: divergent regions in red, conserved in blue, mutation site in C.accent (purple if drug-adjacent). Synchronized rotation: rotating one viewer rotates both.
- **Difference Summary Panel (right, 40%):** Sequence alignment strip showing aligned/misaligned regions. List of top 10 most structurally divergent residue ranges. Per-residue RMSD heat bar (color gradient). Mutation position card (if mutation was specified): shows functional context.

5.5.7 CCA Biomarker Dashboard

⬜ CCA Biomarker Dashboard — /cca

End-to-end clinical demonstration view — TCGA → ranked biomarker candidates.

- **Page Header:** Title: 'Cholangiocarcinoma Biomarker & Drug Target Candidates'. Subtitle: 'TCGA-CHOL · Tumor vs. Normal · [Last updated date]'. Export button (top right): 'Export Full Report (PDF)'.
- **Filter Controls Bar:** Fold-change cutoff slider (default: 2.0×). P-value cutoff input (default: 0.05). Expression direction: Both / Up-regulated / Down-regulated. Analysis type: All / Function-annotated / Drug-targetable. 'Apply Filters' + 'Reset' buttons. Results count badge: 'Showing 47 candidates'.
- **Expression Heatmap (top section, full width):** TCGA expression heatmap — rows: filtered proteins, columns: sample groups (tumor / normal / subtypes). Color scale: blue (low expression) → white (median) → red (high expression). Click a row → scrolls to that protein's detail in ranking table.
- **Candidate Ranking Table (full width):** Columns: Rank · UniProt ID · Gene Name · Fold Change · P-value · Top GO Function · Essentiality Score · Drug Relevance · Evidence Score. Row click → expands inline detail panel showing full ProtMind pipeline report for that protein. Rows color-coded: green left border = drug-targetable; purple left border = known CCA-associated.
- **Detail Expansion Panel (inline, per row):** Function prediction card · PPI summary · Mutation flags · Essentiality score · DrugBank hits. 'View Full Report' link opens /reports/:id. 'Add to Shortlist' button for researcher-curated prioritization.

5.5.8 Admin Dashboard

⬜ Admin Dashboard — /admin

Platform administration: pipeline control, user management, system health monitoring.

- **Left Admin Nav (200px):** Links: Overview · Pipeline Management · User Management · Audit Logs · System Health. Active state: C.blue left border + light blue fill.
- **Overview Panel:** 4 metric cards: Total Proteins in DB · Pipeline Status (last run + next run) · Active Users (last 24h) · Model Versions deployed. Alert banner if any pipeline step failed.
- **Pipeline Management Panel:** Pipeline run history table: Run ID · Start time · Status (badge) · Records processed · Errors. 'Trigger Manual Run' button (primary). Per-step status accordion: Data Ingestion · GO Annotation Filter · Split Computation · Data Card Generation.
- **User Management Panel:** User table: Email · Role · Last active · Status (active/suspended). Role assignment dropdown. Deactivation toggle. Audit log link per user.
- **System Health Panel:** Prometheus metrics embedded: API response time P95 · Error rate · PostgreSQL connection pool · Redis cache hit rate · Object storage usage. Alerting thresholds listed per metric.

5.6 User Flow Diagrams

5.6.1 Primary Flow: First-Time Researcher — Protein Function Query

Step	Screen	User Action	System Response
1	Landing Dashboard	Types 'P01308' in search bar	Autocomplete suggests 'P01308 — Insulin (INS_HUMAN)'
2	Landing Dashboard	Selects suggestion, presses Enter	Routes to /protein/P01308 — protein overview loads
3	Protein Overview	Clicks 'Predict Function'	Routes to /predict/function?id=P01308; job starts
4	Function Prediction	Sees ProgressJob indicator: 'Retrieving embeddings...'	WebSocket pushes progress updates in real-time
5	Function Prediction	Prediction completes — report renders	Summary card shows: 'Hormone activity (GO:0005179) — 94% confidence'
6	Function Prediction	Clicks 'Sequence View' tab	SequenceAnnotator loads with attention heatmap overlay
7	Function Prediction	Hovers residue A14	Tooltip: 'Ala-14 · Attention: 0.87 · Active site (UniProt annotation)'
8	Function Prediction	Clicks 'Evidence' tab	EvidenceGraph renders: prediction → UniProt feature nodes → GO term → 3 literature references
9	Function Prediction	Clicks 'Export PDF'	PDF report generated with all tabs' content, references, and model metadata

5.6.2 Secondary Flow: Clinical Scientist — CCA Candidate Review

Step	Screen	User Action	System Response
1	Landing Dashboard	Clicks 'CCA Biomarker Dashboard' tile	Routes to /cca; expression heatmap and candidate table load
2	CCA Dashboard	Sets fold-change to 3.0×, p-value to 0.01, Direction: Up-regulated	Table filters to 23 candidates; heatmap updates
3	CCA Dashboard	Clicks candidate row: FGFR2 (P21802)	Inline expansion shows: Function = Receptor tyrosine kinase activity · Essentiality: High · Drug hits: 3 (FGFR inhibitors in Phase 2/3)
4	CCA Dashboard	Clicks 'View Full Report'	Opens /reports/FGFR2-CCA with complete pipeline report
5	Report View	Reviews structure, interactions, mutation flags	Sees FGFR2 structural comparison with common activating mutation K650M
6	Report View	Clicks 'Export PDF'	Structured PDF with executive summary, evidence graph, and references generated

5.6.3 Tool-Access Flow: Researcher Using Structure Comparator

Step	Screen	User Action	System Response
1	Landing Dashboard	Clicks 'Structure Comparator' tile	Routes to /tools/structure-comparator
2	Comparator	Enters P01308 in Structure A, P01315 in Structure B	Both proteins resolved: Insulin Human · Insulin Mouse
3	Comparator	Clicks 'Compare'	Both structures fetched from AlphaFold DB; superposition computed
4	Comparator	Sees RMSD = 0.38Å — 'Nearly Identical' badge (green)	Overlaid view shows conserved regions (blue), minor differences (amber)
5	Comparator	Switches to 'Side-by-Side' toggle	Two synchronized Mol* viewers render; rotating one rotates both
6	Comparator	Clicks divergent region in sequence alignment strip	Corresponding residues highlighted in both 3D viewers simultaneously

5.7 Interaction Design

5.7.1 Micro-Interactions

Trigger	Animation	Duration	Purpose
Button click	Scale 0.97 → 1.0 on mousedown release	80ms	Tactile feedback without delay
Card hover	--shadow-sm → --shadow-md, border-top 2px C.mid	150ms ease-out	Affordance — indicates clickability
Panel expand/collapse	Height animation 0 → auto using CSS Grid trick	200ms ease-in-out	Spatial continuity — where did content come from?
Score reveal	Number counter: 0% → actual%, with color fill animating	600ms ease-out	Directs attention; makes confidence legible
Tab switch	Content fade-out 100ms → fade-in 150ms; underline slides	250ms total	Orientation — prevents jarring content swap
Loading skeleton	Gradient shimmer left → right at 1.2s repeat	Continuous	Communicates 'something is coming' without anxiety
Progress step complete	Checkmark stamp animation (scale 0 → 1.1 → 1)	300ms	Closure — confirms pipeline step completion
3D structure loaded	Fade in from transparent + subtle scale 0.98 → 1	400ms	Smooth reveal — avoids jarring pop-in

5.7.2 Keyboard Navigation & Shortcuts

Shortcut	Action
/ (forward slash)	Focus the global search bar from anywhere
Escape	Clear search, dismiss modal, collapse expanded panel
Tab / Shift+Tab	Full keyboard navigation through all interactive elements
Enter	Submit search, confirm action in focused button/link
Arrow keys (in 3D viewer)	Rotate structure by 5° per key press
Ctrl+E / Cmd+E	Export current view/report to PDF

Ctrl+[/ Ctrl+]	Switch between tabs in tabbed views
Ctrl+R / Cmd+R	Reset 3D viewer to default orientation
Number keys 1–6 (on landing)	Quick-open tool tiles 1 through 6

5.7.3 Error & Empty States

Scenario	Visual Treatment	Message Copy
Invalid UniProt ID	Red input border + inline error icon	'This ID doesn't appear in UniProt Swiss-Prot. Please verify it at uniprot.org.'
AlphaFold structure not found	Amber callout	'No AlphaFold structure found. A homology model is being built via SWISS-MODEL — this may take 30–90 seconds.'
Model inference timeout	Red callout + retry button	'Prediction timed out. This may be due to high server load. Your result will be saved when complete.'
Empty PPI network	Neutral empty state illustration	'No interaction data found for this protein in BioGRID or STRING. This may be an understudied protein.'
No CCA candidates at filter threshold	Amber empty state + suggestion	'No candidates meet your current thresholds. Try relaxing the fold-change cutoff or including down-regulated proteins.'
WebGL not supported	Full-page notice	'3D visualization requires WebGL, which isn't available in your current browser. Please use Chrome 90+, Firefox 85+, or Safari 14+.'
API rate limit exceeded	Amber banner (top of page)	'External database rate limit reached. Your request has been queued and will complete automatically.'

5.8 Accessibility (A11y) Standards

Accessibility Commitment

ProtMind targets WCAG 2.1 Level AA compliance across all views. Scientific tools have historically had poor accessibility. ProtMind is designed to be usable by researchers with visual impairments, motor limitations, or assistive technology dependencies.

5.8.1 Contrast & Color

- All body text must meet a minimum 4.5:1 contrast ratio against its background (WCAG 2.1 AA).
- Large text (18px+ bold or 24px+ regular) must meet a minimum 3:1 contrast ratio.
- Color must never be the ONLY means of conveying information — confidence levels must also show numeric values and icons.
- The pLDDT color scale must include pattern fills (hatching) or numeric labels as an alternative for color-blind users. A deuteranopia-safe alternative palette is provided.

5.8.2 Keyboard & Focus

- All interactive elements must be reachable via Tab key navigation.
- Focus ring must be visible on all interactive elements — never suppressed with outline: none without a custom visible alternative.
- The 3D structure viewer (Mol*) exposes keyboard rotation and selection controls per the Mol* accessibility guidelines.
- Skip-to-main-content link must be present and visible on focus at the top of every page.

5.8.3 Screen Reader Support

- All images and icons must have descriptive aria-label or alt attributes.
- Dynamic content updates (WebSocket progress, new prediction results) must use aria-live='polite' regions.
- All data tables must have proper scope attributes on headers and descriptive captions.
- Modal dialogs must trap focus and announce their title via aria-labelledby on open.
- The SequenceAnnotator must expose residue data to screen readers via a structured aria-describedby pattern.

5.8.4 Reduced Motion

- All animations must respect prefers-reduced-motion media query — replace transitions with instant state changes when enabled.
- The 3D structure viewer auto-rotation feature must be disabled by default and only enabled on explicit user action.

5.9 Responsive Design Rules

5.9.1 Desktop (1200px+) — Primary Target

- Full 12-column grid active. All panels visible simultaneously (sidebar + main + context panel).
- 3D viewers render at full resolution with all toolbars visible.
- DataTable shows all columns. Charts render at full size.

5.9.2 Tablet (768–1199px)

- Right context panel collapses into a bottom drawer, accessible via a toggle button.
- Tool tiles grid switches to 2-column layout.
- SplitPanel components default to stacked (top/bottom) instead of side-by-side.
- PAE plot in structure viewer moves below the 3D canvas.

5.9.3 Mobile (<768px)

- Left sidebar collapses to hamburger menu.
- 3D structure viewer renders in simplified mode — controls moved to a bottom action sheet.
- DataTables use horizontal scroll — no column hiding (all data must remain accessible).
- CCA Dashboard heatmap replaces full matrix with a simplified top-20 ranked list view.
- Network graph viewer falls back to a searchable adjacency table on mobile.

5.10 Design Tokens Reference

All design values are stored as CSS custom properties (variables) in a single design token file, enabling consistent theming and future dark mode support.

Token	Value	Category
--color-primary	#1B4F8A	Color — Brand
--color-secondary	#2E75B6	Color — Brand
--color-accent	#0070C0	Color — Brand
--color-bg-light	#EBF3FB	Color — Surface
--color-text-body	#1A1A2E	Color — Text
--color-text-meta	#666666	Color — Text
--color-success	#00B050	Color — Semantic
--color-warning	#FF9900	Color — Semantic
--color-danger	#C00000	Color — Semantic
--color-pharma	#7030A0	Color — Domain
--font-primary	'Inter', sans-serif	Typography
--font-mono	'JetBrains Mono', monospace	Typography
--font-size-base	15px	Typography
--font-size-sm	13px	Typography

--font-size-lg	18px	Typography
--space-unit	4px	Spacing
--radius-default	8px	Shape
--shadow-default	0 1px 3px rgba(0,0,0,0.08)	Elevation
--transition-fast	150ms ease-out	Motion
--transition-base	250ms ease-in-out	Motion
--max-content-width	1440px	Layout
--sidebar-width	260px	Layout
--context-panel-width	300px	Layout

5.11 Design Handoff Checklist

The following checklist must be completed before any design is handed off to the development team.

Item	Owner	Status
All 8 major views have Figma frames at 1440px desktop breakpoint	UI Designer	Required before Phase 2 dev start
All 8 views have mobile (375px) frames	UI Designer	Required before Phase 2 dev start
Color styles defined in Figma matching design token file	UI Designer	Required before Phase 2 dev start
Typography styles defined in Figma matching spec	UI Designer	Required before Phase 2 dev start
All components have all states designed (default/hover/active/loading/error/empty)	UI Designer	Required per component
All icons are exported as SVG from Figma and added to the component library	UI Designer	Required per icon
Prototype flows built in Figma for the 3 primary user flows (section 5.6)	UI Designer	Required for stakeholder review
Contrast ratios verified for all text/background combinations (A11y section 5.8.1)	UI Designer + QA	Required before development

Component library stories added in Storybook for every component in section 5.3	Frontend Engineer	Required before integration
Design tokens exported to design-tokens.css and committed to the repository	Frontend Engineer	Required before any CSS is written
3D viewer color scheme (pLDDT) verified to match AlphaFold DB exactly	Frontend Engineer	Required for Phase 2 completion

Chapter 5 End

This UI/UX Design chapter constitutes the complete design specification for all ProtMind views, components, and interaction patterns. All wireframe specifications, design tokens, component properties, and accessibility requirements documented here are the authoritative source of truth for frontend development across all phases.